Stereoisomerism — a complete handwritten short-note set

Part I is the JEE Main and JEE Advanced core: geometrical, optical and conformational isomerism, the counting rules, and the reaction stereochemistry JEE actually asks. Part II is the JEE-Advanced-level extension — symmetry and S_n, general counting, ring junctions, quantitative conformational analysis, reaction design.

What is inside

Section	Sheets	Theme
1. Foundations	J-01 – J-02	Isomerism map, vocabulary, why stereochemistry is tested
2. Geometrical	J-03 – J-09	Conditions, E/Z, rings, cumulenes, counting, physical properties
3. Optical	J-10 – J-20	Polarimetry, chirality, symmetry, meso, projections, R/S, counting
4. Conformational	J-21 – J-24	Barriers, ethane and butane, cyclohexane
5. Reactions	J-25 – J-27	S_N1/S_N2, E_2, addition, worked enumeration
6. Exam craft	J-28 – J-29	Twenty traps, one-page recall

Table 0.1 — the route map for Part I.

How to read a page

Each sheet has one argument: a definition, the reason behind it, a worked example, then the result to remember and the trap that follows from it. The colour carries the meaning, so read the colour before the words.

- **Navy handwriting** — the explanation and the working.
- **Teal headings and teal panels** — the reason something is true (why it works).
- **Yellow panels** — KEY RESULT: the line to memorise, or the formula to reproduce.
- **Red panels** — TRAP: the mistake that examiners build options around.

KEY RESULT
If you remember only one sentence from the whole set, remember this: **stereochemistry is decided by the 3-D drawing, not by the name of the functional group.** Every line here is a shortcut for a question you could answer by drawing carefully.

Source and scope of these notes

Built from the supplied 57-lecture note file (NS Sir, Stereoisomerism) and checked against the NCERT Class XI Unit 10 and Class XII Unit 10 stereochemistry content; Part II adds the JEE Advanced statements of the same topics and nothing beyond. Where the source and a standard text disagreed, the correction is recorded in SOURCE-AUDIT.md and the corrected statement is taught here.

NOTE
These are study notes, not a textbook. Values marked ≈ are typical textbook values quoted to fix the order of magnitude; they are not measurements made for this set. Every structure was drawn and checked by hand-encoded rules, but no claim of exhaustive chemical verification is made.

Printing

A_4 portrait, 10 mm margins, nominal 300 dpi; print at 100 % (Fit, not Fill or Crop), one sheet per page. The PNGs are 2480 × 3508 px. There is no Olympiad section: Part II stops at JEE Advanced.

Start every problem from the molecular formula. If two structures differ in which atom is joined to which, they are **structural** isomers; if the connectivity is identical and only the arrangement in space differs, they are **stereoisomers**. The whole chapter lives in the right-hand branch of the map below.

1.1 The isomerism tree

Level	The question to ask	Examples
Structural (constitutional)	Is the connectivity different?	butane / 2-methylpropane; propan-1-ol / methoxyethane
chain, position, functional	which carbon, which functional group?	pentane isomers; butan-1-ol / butan-2-ol; ethanol / dimethyl ether
Stereoisomer	same connectivity, different 3-D arrangement	see the next two rows
Conformational	can it interconvert without bond breaking?	anti/gauche butane; chair/boat cyclohexane
Configurational	does it interconvert by rotation?	cis/trans but-2-ene; (R)/(S) lactic acid

Table 1.1 — the classification used throughout these notes. Structural is the left branch; stereoisomerism is everything below.

Configurational isomers are further split into **geometrical** (cis/trans, E/Z, syn/anti) and **optical** (enantiomers and diastereomers, including meso forms). Conformational isomers are the subject of section 4.

1.2 Six words that must be exact

1. **Stereocentre** — an atom bearing groups arranged so that interchanging two of them gives a stereoisomer. For carbon this usually means four different groups; it is a property of one atom.
2. **Chiral** — the molecule is not superimposable on its mirror image. It is a property of the whole molecule, and it is the condition for optical activity.
3. **Enantiomers** — a pair of stereoisomers that are non-superimposable mirror images. Identical physical properties, opposite rotation.
4. **Diastereomers** — stereoisomers that are not mirror images. Different physical properties, separately isolable by ordinary means.
5. **Meso** — a molecule with stereocentres that is nevertheless achiral because an internal plane (or centre) makes its mirror image superimposable.
6. **Racemic** — a 1:1 mixture of two enantiomers, optically inactive by external compensation.

KEY RESULT
Chirality is decided by the molecule, not by the count of stereocentres. Meso compounds have stereocentres and are achiral; allenes and biphenyls can be chiral with no stereocentre at all. Both directions are examinable.

TRAP
"Optically inactive" is **not** a synonym for "achiral". A racemic mixture is optically inactive although both of its components are chiral. Quote the reason for inactivity: no chiral arrangement, internal compensation, or external compensation.

1.3 Why the topic is worth marks

Stereochemistry is the part of organic chemistry where a reaction can be right and still give the wrong answer. S_N2 gives inversion; S_N1 gives racemisation; syn addition to an alkene gives a meso product from one face and a racemic pair from the opposite faces; hydrogenation of an optically active alkene can destroy the activity. JEE Advanced has repeatedly fused this chapter with S_N1/S_N2, addition and hydrogenation rather than asking a definition directly.

The drug thalidomide is the standard illustration: the (R) enantiomer is the sedative, the (S) enantiomer is teratogenic, and under the racemisation conditions in vivo the two interconvert through the acidic C–H next to the imide. The point is chemical, not pharmacological: a stereocentre next to an acidic C–H is not configurationally stable.

1.4 How the examiner phrases it

Question stem	First move	What earns the mark
"How many stereoisomers?"	count skeletons, then centres	the number and the split into active/meso
"Are they identical, enantiomers or diastereomers?"	normalise both to one projection	the evidence (centres inverted or not)
"Is the compound optically active?"	look for σ, i, S_n	the symmetry element, not the number of centres
"Assign R/S"	rank by CIP, put the lowest group away	the priorities written out
"Which is more stable?"	count gauche/eclipsing or a/e interactions	the interaction that decides it
"Predict the product stereochemistry"	name the mechanism first	the mechanism → inversion / racemisation / syn / anti

Table 1.2 — the six stems that cover almost every objective question on this chapter.

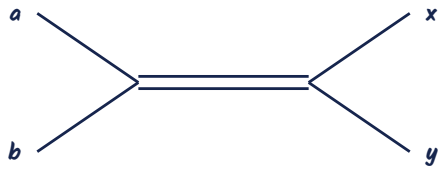
Two habits save marks. First, when the stem says "stereoisomers", give the total **and** the number of optically active forms; most marking schemes award the second number separately. Second, when the stem says "optically active", do not count centres — look for the plane. A molecule with two centres and a plane is inactive, and a molecule with no centre and no plane (an allene) is active.

A third habit is worth the ink: write the molecular formula before anything else. $C_4H_8Cl_2$ tells you the skeleton is saturated, so any ring you draw is a real possibility; C_5H_8 tells you two degrees of unsaturation, so a ring plus a double bond is possible. Enumerations fail more often from a missed skeleton than from a missed stereocentre.

2.1 Why a C=C freezes the arrangement

The π bond is built from two p orbitals that overlap side-on. Twist the double bond by 90° and the overlap falls to zero, so the π bond must break before the two ends can exchange places. The barrier is about **250–270 kJ mol⁻¹**, roughly twenty times the thermal energy available at room temperature; a single C–C bond needs only 12–20 kJ mol⁻¹ and rotates freely at 25 °C.

$$E(\pi,\ C{=}C) \approx 250\text{-}270\ \text{kJ mol}^{-1}$$
$$E(\sigma,\ C\text{-}C) \approx 12\text{-}20\ \text{kJ mol}^{-1}$$



the π bond stops rotation

a ≠ b and x ≠ y

→ geometrical isomers (E/Z)

Fig. 2.1 — the two conditions are independent. Restricted rotation without two different groups at both ends gives no isomerism; two different groups without restricted rotation gives only conformers.

2.2 Condition 1 — restricted rotation

Rotation must be blocked in a way that survives at room temperature. Three situations qualify: a π bond (C=C, C=N, N=N), a ring (the two substituents are locked above and below the mean plane), and a sterically locked single bond (the atropisomerism of section 6.11 in Part II, where the barrier is high enough that the conformers can be isolated). A single bond that rotates freely never gives geometrical isomers, however different the two drawings look.

2.3 Condition 2 — two different groups at each end

On the left carbon, call the groups a and b; on the right, x and y. Isomerism exists only if **a ≠ b and x ≠ y**. This is why terminal alkenes ($CH_2{=}CHR$) and geminally disubstituted alkenes ($(CH_3)_2C{=}CH_2$) have no geometrical isomerism: one carbon carries two identical atoms, so interchanging the two ends produces the same molecule.

Structure	Left end	Right end	Geometrical?
$CH_3CH{=}CHCH_3$	CH_3, H	CH_3, H	yes — cis and trans (2)
$CH_2{=}CHCH_3$	H, H	—	no (identical H on one carbon)
$(CH_3)_2C{=}CHCH_3$	CH_3, CH_3	CH_3, H	no (two CH_3 on one carbon)
$CHCl{=}CHBr$	H, Cl	H, Br	yes — and cis/trans cannot name it; use E/Z

Table 2.1 — apply the test to each end separately.

KEY RESULT

Two questions, two answers, one conclusion: **is rotation blocked?** and **are both ends unsymmetrical?** Only two yeses give a geometrical centre.

TRAP
Do **not** count a C=C twice. One double bond is one geometrical centre, giving at most two isomers. The count 2^n uses n = number of centres, not number of atoms.

2.4 The four kinds of centre

1. **C=C** in an open chain or a large ring; each sp^2 carbon planar, each end unsymmetrical.
2. **C=N** in oximes and imines: nitrogen is sp^2 and carries a lone pair, so the nitrogen end is always unsymmetrical — only the carbon end needs checking.
3. **N=N** in azo compounds and diazoates: each nitrogen carries one organic group plus a lone pair.
4. **Ring carbons** of cycloalkanes: substituents are locked above/below the ring; the ring itself supplies the restricted rotation.

For a C=N centre the nitrogen lone pair occupies the third sp^2 orbital. That lone pair counts as a substituent both for the isomerism test and for naming (E/Z or syn/anti). The same is true of the N=N centre.

2.5 How oximes are made (the reaction behind the question)

$R_2C{=}O + NH_2OH \rightarrow R_2C{=}N{-}OH + H_2O$ (mild acid, pH 4–5). Aldehydes give aldoximes (RCH=NOH); ketones give ketoximes ($R_2C{=}NOH$). The reaction is reversible, so an oxime can be hydrolysed back to the carbonyl compound, and hydroxylamine (or hydrazine) is the standard test that distinguishes a carbonyl from an alcohol or ether.

Carbonyl compound	Oxime stereoisomers	Reason
HCHO	1	carbon bears two H
RCHO (x ≠ y)	2	carbon bears H and R
$R_2C{=}O$ with R ≠ R′	2	two different groups
$R_2C{=}O$ with R = R′	1	two identical groups
a diketone, two C=N formed	up to 4	each centre doubles the count; check symmetry

Table 2.2 — the standard "number of products" question.

TRAP
For a symmetric diketone such as butane-2,3-dione the two C=N centres are equivalent; syn,syn and anti,anti are two distinct compounds, but syn,anti is one compound, not two. Enumerate, then merge by symmetry.

2.6 The C=N family: hydrazones, imines and semicarbazones

Every reaction that condenses a carbonyl with an amine derivative makes a C=N centre, and every one of them inherits the same stereochemistry. The table below is worth learning as a family, because the question "how many isomers?" is identical for all four.

Reagent	Product	C=N formed	Stereoisomers for RCHO
NH_2OH	oxime RCH=N–OH	yes	2 (syn/anti, or E/Z)
NH_2NH_2	hydrazone $RCH{=}N{-}NH_2$	yes	2
NH_2NHR	substituted hydrazone	yes	2
NH_2CONH_2	semicarbazone $RCH{=}N{-}NHCONH_2$	yes	2
$NH_2C(=NH)NH_2$	guanidine condensate	yes	2

Table 2.3 — all five families create the same kind of centre, so the counting rule is shared.

For a ketone R_1COR_2 the count depends on whether the two groups differ: $R_1 \neq R_2$ gives two isomers, $R_1 = R_2$ gives one. For a diketone the centres are counted independently and then merged by symmetry — butane-2,3-dione gives three (syn,syn; anti,anti; syn,anti), not four.

The nitrogen substituent matters for naming but not for counting. In an oxime the -OH is the reference group; in a hydrazone the $-NH_2$ is. E/Z is assigned from the CIP priorities in each case, with the nitrogen lone pair always lowest.

KEY RESULT
Carbonyl + any NH_2-X reagent → C=N centre. The number of stereoisomers depends only on the two groups on the carbonyl carbon, never on what X is.

TRAP
Do not write "oximes show cis/trans". The correct descriptors are syn/anti (with the reference group named) or E/Z. cis/trans needs a pair of comparable groups at both ends, and the nitrogen end carries a lone pair rather than a second carbon group.

2.6b Oxime chemistry beyond the count

The C=N centre is not only a counting problem; it is a functional group with two examinable reactions. Both follow from the same geometry: the nitrogen lone pair fixes a plane, so the groups on the carbon are locked on one side or the other.

1. **Formation** is acid-catalysed and reversible: the amine attacks the carbonyl, a carbinolamine forms, and dehydration gives C=N. Because every step is reversible, the more stable (usually E) oxime accumulates — the reaction is under thermodynamic control.
2. **Hydrolysis** reverses it: aqueous acid regenerates the carbonyl and hydroxylamine. This is why oximes are used as protecting groups for aldehydes and ketones.
3. **Beckmann rearrangement** (H_2SO_4 or PCl_5): the group anti to the -OH migrates to nitrogen and the product is a substituted amide, RCONHR′. The migrating group keeps its configuration, so the reaction is stereospecific — cyclohexanone oxime gives caprolactam, the monomer for nylon-6.

The Beckmann point is worth stating precisely: it is the group **anti** (trans) to the hydroxyl that migrates, not the group syn to it. For a ketoxime with two different groups, that means the two geometrical isomers give two different amides — a clean demonstration that E/Z isomerism is chemically consequential.

Worked E/Z assignment: butan-2-one oxime, $CH_3C(=NOH)CH_2CH_3$. On the carbon, ethyl (C,C,H) beats methyl (C,H,H), so ethyl is priority 1. On nitrogen, -OH beats the lone pair, so -OH is priority 1. If the -OH is on the same side as the ethyl group the oxime is (Z); on the opposite side it is (E). Both are isolable compounds, and their Beckmann products differ.

The 2,4-DNP test. 2,4-Dinitrophenylhydrazine condenses with aldehydes and ketones to give a yellow-orange precipitate of the hydrazone. The test detects the C=O group, and the product it forms is exactly the C=N centre of the table above; a positive test therefore also means the product can exist as E/Z isomers.

KEY RESULT
For a C=N centre: count the isomers from the carbon substituents (two different groups → 2), name them E/Z from CIP priorities, and remember that the two isomers can react differently (Beckmann).

TRAP
Do not write that oximes are "stable" or "unreactive". They hydrolyse in aqueous acid and rearrange in concentrated acid — both reactions are used in synthesis and both are examinable as applications of the C=N geometry.

cis/trans works only when each end of the double bond carries a pair of comparable groups. When the four groups are all different — CHCl=CHBr, for example — there is no "same side" pair to name. The universal descriptor is E/Z, built on the Cahn-Ingold-Prelog (CIP) priority rules.

2.7 E and Z

Assign priority 1 and 2 at the left carbon, then priority 1 and 2 at the right carbon. If the two priority-1 groups lie on the **same side** of the double bond the descriptor is **Z** (zusammen, together); if they lie on **opposite sides** it is **E** (entgegen, opposite).

KEY RESULT
E/Z compares priority-1 groups, not "like" groups. Two different alkenes can both be (E) even though their substituents are not the ones you would call cis/trans.

TRAP
Priority 1 groups together = E. Priority 1 groups opposite = Z. This is the single most common slip; say the rule aloud once before you assign.

2.8 The CIP rules, in the order you apply them

1. **Higher atomic number at the first atom.** I > Br > Cl > S > O > N > C > H. If the first atoms differ, stop here.
2. **Compare the next layer.** List the atoms attached to the first atom in decreasing atomic number and compare the lists position by position (highest first). Do not jump to a deeper layer while the present layer is still undecided.
3. **Same element, higher mass number wins.** D > H, T > D.
4. **Lone pair is the lowest possible ligand.** A real atom to any atom always beats a lone pair.
5. **Multiple bonds are expanded.** Each double bond counts as two single bonds to that atom; a triple bond counts as three. C=O is C bonded to (O, O, phantom O); C≡N is C bonded to (N, N, phantom N). This is bookkeeping, not a real structure.

Group	Layer 1	Layer 2 (as written)	Verdict
$-CH_3$	C	H, H, H	lowest carbon group
$-CH_2Cl$	C	Cl, H, H	beats $-CH_3$ at layer 2
$-CH_2OH$	C	O, H, H	below $-CH_2Cl$ (O loses to Cl at layer 2)
$-CHO$	C	O, O, H (one O phantom)	beats $-CH_2OH$
$-COOH$	C	O, O, O (one O phantom, one OH)	beats $-CHO$
$-C\equiv N$	C	N, N, N (two phantom N)	below $-COOH$ (O beats N at layer 2)
$-C\equiv CH$	C	C, C, H (two phantom C)	ties with $-CH=CH_2$ (same expansion)

Table 2.4 — expand multiple bonds before comparing anything to the right.

TRAP
The commonest E/Z error is comparing the second atom of a group before checking the first atom of a multiple bond. Expand every multiple bond into phantom atoms **first**, then compare layer by layer.

2.9 Worked examples

(a) 1-bromo-1-chloropropene, $CH_3CH=CBrCl$. At the left carbon CH_3 beats H. At the right, Br beats Cl. If the CH_3 and Br are on the same side, the compound is (Z)-1-bromo-1-chloroprop-1-ene; if opposite, (E). Note that "cis" would be meaningless here.

(b) But-2-enedioic acid, HOOC-CH=CH-COOH. Both ends carry H and COOH, so cis/trans is available (maleic = cis, fumaric = trans). The priority groups are COOH at both ends, so maleic acid is (Z) and fumaric acid is (E). This pair is worth memorising because it reappears in the physical-property questions.

(c) An oxime. For $R_2C=N-OH$, the nitrogen ligands are -OH and the lone pair; oxygen beats the lone pair, so -OH is priority 1 on nitrogen. On carbon the higher-priority carbon group is priority 1. If the two priority-1 groups are on opposite sides the oxime is (E).

(d) A diene with two centres. Each double bond gets its own descriptor and its own locant: (2E,4Z)-hexa-2,4-diene. Do not average or combine them.

2.10 cis/trans, syn/anti, E/Z — which name to use

Descriptor	Applies to	Requirement	Notes
cis / trans	C=C, C=N rings	a comparable pair of groups at the two ends	relative, not universal
syn / anti	C=N, N=N	the OH (or group) compared with a named group	always quote the reference group
E / Z	any geometrical centre	CIP priorities only	universal; use it when in doubt

Table 2.5 — use E/Z for any centre with four different groups.

> NOTE
> Indian textbooks use "syn" for oximes in two different senses: sometimes "OH on the same side as H" and sometimes "OH on the same side as the named R". Because the two conventions coexist, JEE has moved to E/Z for oximes. If a question uses syn/anti, state the reference group explicitly in your answer.

2.11 Four CIP drills with the reasoning written out

(a) $-CH(Br)CH_2Br$ versus $-CH_2CH_2Br$. Both start with C. Layer 2: (Br, H, C) versus (C, H, H). Br wins at the first comparison, so $-CH(Br)CH_2Br$ is the higher priority group; the longer chain is irrelevant because the decision was made at layer 2.

(b) $-CH(OH)CH_3$ versus $-CH(OH)CH_2OH$. Layer 1: C versus C. Layer 2: (O, H, C) versus (O, H, C) — still tied. Layer 3: the first group has (H, H, H) on its carbon, the second has (O, H, H), so the second group wins. This is how the aldehyde end of a sugar is ranked above the alcohol end.

(c) -CHO versus -COOH. Expand the multiple bonds. -CHO is C(O, O, H) because the C=O counts twice; -COOH is C(O, O, O) because the C=O counts twice and the OH adds one O. Compare position by position: both have O, O; the third ligand decides, O > H, so -COOH wins.

(d) $-C\equiv N$ versus $-C\equiv CH$. Expand: (N, N, N) versus (C, C, H). N beats C at the first position, so $-C\equiv N$ wins. Note that the two groups have the same formula weight — the decision is about atoms, never about mass.

Comparison	Decided at layer	Winner
-Br vs -Cl	1	-Br
-OH vs $-NH_2$	1	-OH
$-CH_2Cl$ vs $-CH_3$	2	$-CH_2Cl$
$-CH(CH_3)_2$ vs $-CH_2CH_2Br$	2	$-CH(CH_3)_2$
-CHO vs $-CH_2OH$	2	-CHO
-COOH vs -CHO	3	-COOH
-C≡N vs -C≡CH	3	-C≡N (N beats C)
-D vs -H	mass	-D

Table 2.6 — check which layer settles each comparison before comparing further.

TRAP
The single most expensive mistake in this table is comparing a later atom of one group with an earlier atom of the other. Compare layer by layer, position by position; a bromine three bonds away never beats an oxygen one bond away.

2.11b E/Z in rings and in polyenes

In a ring, "cis" and "trans" are still usable because the ring provides a reference plane, but the moment three or more substituents are present, cis/trans stops identifying the compound uniquely and E/Z is safer. E/Z in a ring is assigned exactly as in an open chain: compare the two groups on each alkene carbon by CIP priority, then look at which side of the double bond the priority-1 groups lie on.

For a polyene, every double bond is assigned independently and the descriptors are listed with locants in ascending order: (2E,4Z,6E)-octa-2,4,6-triene. Do not average them and do not let one centre's assignment influence the next. The only coupling is through the chain geometry: a 2,4-diene drawn all-trans is a zig-zag, while a 2E,4Z diene has a pronounced kink, which is why the two differ in melting point and in their cyclisation behaviour.

Structure	Descriptors	Why it matters
(2E,4Z)-hexa-2,4-diene	E at C_2, Z at C_4	kinked; different shape from all-E
(2E,4E)-hexa-2,4-diene	E, E	zig-zag; the more stable isomer
(1Z,3Z)-cycloocta-1,3-diene	Z, Z	small-ring dienes are forced cis
(2E)-3-methylpent-2-ene	E	CH_3 vs CH_2CH_3 decides, not chain length

Table 2.7 — descriptors are per centre, listed with locants.

KEY RESULT
E/Z is assigned per double bond, in ascending locant order, and each assignment is independent. In rings, E/Z is safer than cis/trans as soon as more than two substituents are present.

2.12 Why a ring is a geometrical centre

A cycloalkane is not flat for $n \geq 4$, but every ring carbon still has one bond pointing to one face and one to the other. Turning a substituent from "up" to "down" would require the ring to pass through itself, so the up/down arrangement is configurationally locked. That is exactly the situation a C=C provides electronically.

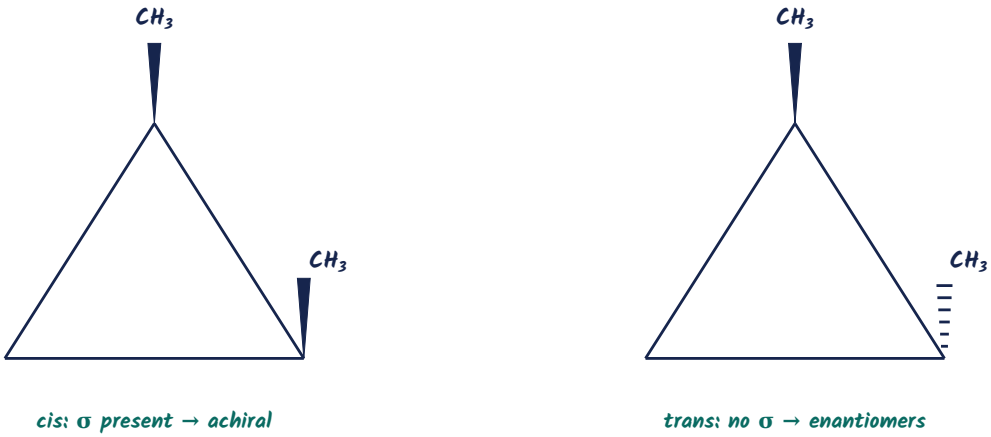


Fig. 2.2 — cis-1,2-dimethylcyclopropane has a mirror plane through C_3 and the C_1-C_2 bond; the trans isomer has none and exists as a pair of enantiomers.

2.13 Counting substituent patterns on a small ring

1. **Anchor one substituent up.** Turning the whole molecule over is allowed, so the all-up and all-down drawings are the same compound.
2. **List the remaining up/down choices**, then delete duplicates produced by (i) rotating the ring and (ii) turning the whole molecule over.
3. **Treat identical substituents as identical** — two Br atoms are not distinguishable, so swapping their labels is not a new isomer.
4. **Test each surviving pattern for σ and i.** A pattern with a mirror plane is one achiral compound; a pattern without one is a pair of enantiomers and contributes 2 to the stereoisomer count.

Ring compound	Geometrical isomers	Total stereoisomers	Which are chiral
1,2-dimethylcyclopropane	2 (cis, trans)	3	trans (pair); cis is meso
1,2-dimethylcyclobutane	2 (all-cis, one trans)	4	the one trans form (pair)
1,3-dimethylcyclobutane (different groups)	3	6	the Br-trans form (pair)
1,2-dimethylcyclopentane	3 (cis, trans, cis′)	4	cis′ is meso; trans has a pair
1,3-dimethylcyclopentane	3	4	trans′ (pair), cis has σ
1,2-dimethylcyclohexane	2	3	trans (pair); cis — both have σ

Table 2.8 — patterns worth recognising. Count geometrical isomers first, then split the chiral ones into pairs.

2.14 Worked enumeration: 1,2-dibromo-3-methylcyclopropane

Put Br at C_1 and C_2, CH_3 at C_3. Anchor Br(1) up. The Br(2) choice splits into same face (up) and opposite face (down). With Br(1) = Br(2) = up, the CH_3 has two choices, but turning the ring over makes them identical, so Br(2) = down with CH_3 up is the same compound as Br(2) = up with CH_3 down (relabel the bromines). That leaves three geometrical arrangements:

- **all three up (or all three down)** — a mirror plane through C_3 and the C_1-C_2 bond → achiral, 1 compound.
- **both Br on one face, CH_3 on the other** — again a mirror plane bisecting the Br-Br bond → achiral, 1 compound.
- **the two Br on opposite faces, CH_3 either up or down** — no plane, no centre → chiral, a pair, 2 compounds.

KEY RESULT
Three geometrical isomers become four stereoisomers: 1 + 1 + 2. Always report both numbers if the question allows it, and say which are optically active.

2.15 Double bonds inside small rings

A C=C built into a ring is forced to be cis: the ring chain leaves the two sp^2 carbons on the same side. A trans-cycloalkene would need the chain to cross to the other face of the double bond, which is geometrically impossible for small rings. The rule of thumb is: rings up to seven members show only the cis alkene; cyclooctene is the first that can accommodate a trans double bond, and trans-cyclooctene is strained but isolable.

Trans-cycloalkenes are also chiral. The twist of the ring destroys every mirror plane, so it exists as two enantiomers that interconvert only slowly. That is a genuine JEE-Advanced-standard point, and it is not "a dangerous half-truth".

TRAP
For a small-ring alkene, "how many geometrical isomers?" has answer 1, not 2. If the question asks for the number of stereoisomers and the compound also has stereocentres, count those separately.

2.16 A harder ring enumeration: 1,2,3,4-tetramethylcyclobutane

Four substituents, all identical, on a four-membered ring. Anchor one methyl at C_1. The remaining three positions have three choices, but the square has a four-fold rotation and a mirror symmetry, so most of the 8 raw patterns collapse.

Pattern (relative to C_1 up)	Distinct?	Symmetry	Stereoisomers
up, up, up, up (all cis)	yes	σ and *i*	1
up, up, up, down	yes	σ	1
up, down, up, down (alternating)	yes	*i*, σ	1
up, down, up, down (alternating, other phase)	same as above	rotation by 60°	—
up, up, down, up	same as up, up, up, down	rotation	—
up, down, down, up	same as up, up, down, down	rotation	—

Table 2.9 — six of the eight patterns reduce to three compounds.

So 1,2,3,4-tetramethylcyclobutane has three stereoisomers, all achiral. The all-cis form has a plane through the ring and a centre; the up-up-up-down form has a plane; the alternating up-down-up-down form has a centre of symmetry. None is chiral, which is a good reminder that "many substituents" does not mean "chiral".

2.17 Why the up/down method always works

The method is a direct application of the definition of a stereoisomer: same connectivity, different spatial arrangement. On a ring, the only spatial variable available to a substituent is which face it occupies, so enumerating faces is a complete enumeration. What must be removed afterwards is duplication by symmetry: ring rotation, reflection, and turning the whole molecule over.

KEY RESULT
Enumerate in three stages: anchor one group → list the up/down words → delete words related by rotation or global flip. Then test each survivor for σ and i to decide chiral or achiral.

2.17b Different substituents on the same ring

When the two substituents are different, the end-for-end equivalence disappears and the count usually rises. Work 1-bromo-2-chlorocyclopropane: anchor Br up. Cl may be up (cis) or down (trans). The cis arrangement has a mirror plane through C_3 and the C_1-C_2 bond, so it is achiral — one compound. The trans arrangement has no plane and no centre, so it is a pair of enantiomers — two compounds. Total: two geometrical isomers, three stereoisomers, two of them active.

Now 1-bromo-1-chloro-2-methylcyclopropane. C_1 carries Br and Cl and is not a stereocentre (the ring paths are different, but the two halogen groups differ, so it is a stereocentre — check: its four groups are Br, Cl, C_2 and C_3, all different, so it is a stereocentre). C_2 carries H, CH_3, C_1 and C_3, also four different groups. Two coupled centres on a ring: cis and trans give 2 geometrical isomers, each chiral (no plane), so 4 stereoisomers in two enantiomeric pairs.

Ring compound	Geometrical	Stereoisomers	Active forms
1-bromo-2-chlorocyclopropane	2	3	2 (trans pair)
1-bromo-1-chloro-2-methylcyclopropane	2	4	4 (two pairs)
1,2-dibromocyclopropane	2	3	2 (trans pair)
1,2,3-tribromocyclopropane	3	4	2 (one pair)

Table 2.10 — with different substituents, cis is often still achiral, but trans is always chiral.

TRAP
Do not use the "all-cis is achiral" shortcut on a ring carrying three different substituents. All-cis-1,2,3-tribromocyclopropane has a plane (it is achiral), but all-cis-1-bromo-2-chloro-3-methylcyclopropane does not — the three groups are different, so no plane can map them onto each other.

2.18 Geometry of a cumulene

A cumulene has consecutive double bonds, C=C=C=... . Each inner carbon is sp hybridised and linear, and its two π bonds use two different p orbitals, which are perpendicular. The consequence is that the substituent planes at the two ends are either **perpendicular** or **coplanar**, depending on how many double bonds lie between them.

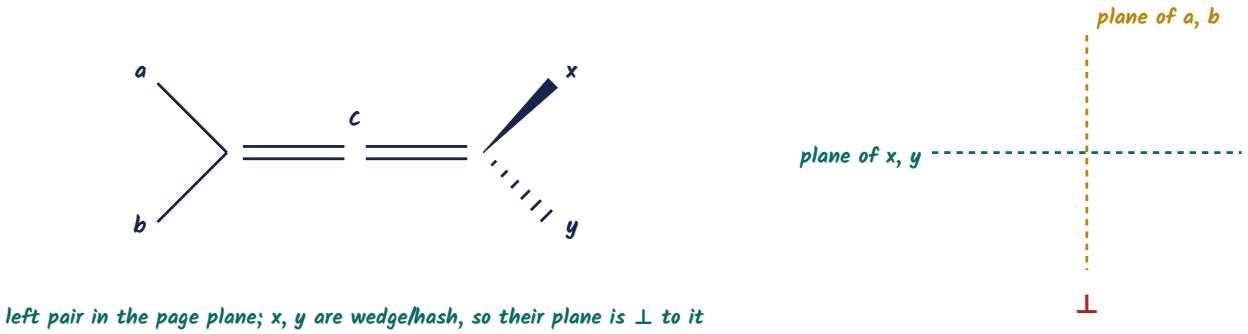


Fig. 2.3 — in an allene the two end planes are perpendicular. Substituting both ends with two different groups each removes every mirror plane and leaves a chiral axis.

Consecutive C=C	Example	End planes	Result (both ends unsymmetrical)
2 (even)	allene, $CH_2=C=CHCH_3$	perpendicular	no cis/trans; chiral (a pair of enantiomers)
3 (odd)	butatriene, $CH_3CH=C=C=CHCH_3$	coplanar	cis/trans (E/Z) pair; achiral
4 (even)	pentatetraene	perpendicular	chiral if both ends differ

Table 2.11 — the odd/even rule. Count the double bonds between the two end carbons, not the atoms.

KEY RESULT
Odd number of consecutive double bonds → coplanar ends → geometrical isomerism. Even number → perpendicular ends → no geometrical isomerism, but a chiral axis if both ends carry two different groups.

2.19 Why perpendicular ends give chirality, not cis/trans

Take an allene with groups a, b at one end and x, y at the other. Because the two end planes are perpendicular, swapping a and b at one end produces a structure that cannot be superimposed on the original — but neither can it be called cis or trans, because there is no common reference plane. The two arrangements are mirror images, and they are not interconvertible without breaking the π system. That is a pair of enantiomers, and the molecule has **axial chirality** with no stereocentre anywhere.

If either end carries two identical groups, a mirror plane appears through the axis and the molecule becomes achiral. So the test is: **both ends unsymmetrical, even number of double bonds → chiral.**

2.20 Rings count as double bonds

A ring also locks a plane, so for the odd/even count **each ring is equivalent to one double bond**. A chain such as ring-C=C-ring therefore has three units (odd) and shows cis/trans-type isomerism; ring=C (an exocyclic double bond on a ring) has two units (even) and behaves like an allene.

- Ring-C=C-ring: 1 + 1 + 1 = 3 units, odd → the two rings are coplanar → geometrical isomers exist.
- Ring=C(R)-R′: 1 + 1 = 2 units, even → perpendicular → no geometrical isomerism.
- Spiro compound: the spiro atom joins two rings in perpendicular planes — the same situation as an allene.

2.21 Spiro compounds

Two rings sharing exactly one carbon lie in **perpendicular planes**, because that carbon uses two bonds for one ring and two for the other, and tetrahedral geometry puts those pairs at right angles. A spiro compound therefore behaves like an allene: no geometrical isomerism, but **axial chirality** if each ring is unsymmetrically substituted. If either ring has a mirror plane of its own through the spiro atom, the whole molecule has a plane and is achiral.

TRAP
Do not call a spiro compound "cis/trans". There is no double bond and no common plane; the correct vocabulary is chiral/achiral, and the stereoisomers are enantiomers.

The classic example is 2,2′,6,6′-tetramethylbiphenyl-3,3′-dicarboxylic acid (the "Mills-Nixon" molecule, whose two enantiomers were resolved in 1933 — the first proof that restricted rotation alone can generate chirality). It sits in the same chapter because the argument is the same one used for allenes: a locked axis, two unsymmetrical ends, no plane.

2.21b Worked cumulene and spiro questions

1. **How many stereoisomers for $CH_3CH=C=CHCH_3$?** Two consecutive C=C (even), end planes perpendicular, each end carries H and CH_3 → a pair of enantiomers, so **2** stereoisomers, both optically active.
2. **How many for $CH_3CH=C=C=CHCH_3$?** Three consecutive C=C (odd), planes coplanar → cis and trans geometrical isomers, each with a plane → **2** stereoisomers, both achiral.
3. **How many for $(CH_3)_2C=C=CH_2$?** One end carries two CH_3 → a mirror plane through the axis → **1** compound, achiral.
4. **How many for 1,3-dimethylallene?** The same as the first case: 2 enantiomers. The name "1,3-dimethylallene" is another way of writing $CH_3CH=C=CHCH_3$.

For spiro compounds the counting is per ring. Spiro[3.3]heptane itself has two identical rings and two planes, so it is achiral. 2,6-dimethylspiro[3.3]heptane, with one methyl on each ring at non-equivalent positions, has no plane and exists as a pair of enantiomers. The test is always the same: draw the two rings in perpendicular planes and look for a mirror plane that contains the spiro atom.

KEY RESULT
Cumulene count: even number of C=C → chiral pair (if both ends differ); odd → cis/trans pair (achiral). Spiro count: test each ring separately, then look for a plane through the spiro atom.

2.22 Two rings sharing atoms

A bicyclic compound has two rings that share at least two atoms. In a fused system the two rings share one bond (decalin); in a bridged system they share three or more atoms (bicyclo[2.2.1]heptane). The shared atoms are the bridgeheads, and the key stereochemical question is whether the two bridgeheads can sit on opposite faces.

At a ring junction each bridgehead carbon carries one hydrogen. If both hydrogens are on the same face the junction is **cis**; if they are on opposite faces it is **trans**. Cis and trans junctions are diastereomers with different energies, and at least one of them changes the shape of the whole framework.

System	cis/trans junction?	Why
bicyclo[2.2.1]heptane (norbornane)	no	the one-carbon bridge is too short to reach the other face
bicyclo[2.2.2]octane	no	bridges still too short; only the cis junction is geometrically possible
decalin (bicyclo[4.4.0]decane)	yes	six-membered rings can close on either face; both isomers are stable
hydrindane (bicyclo[4.3.0]nonane)	yes	one five- and one six-membered ring
large-ring fused systems (≥ 10 atoms)	yes	enough flexibility to place the bridges on opposite faces

Table 2.12 — the size rule is about the whole framework, not about a single ring.

NOTE
A popular shortcut says "a bicyclic compound shows geometrical isomerism only if it contains a ring of eight or more atoms". That is not reliable: decalin shows cis/trans ring junctions with six-membered rings. Decide by asking whether the two bridges can reach opposite faces, and by drawing.

2.23 cis- and trans-decalin

In cis-decalin the two junction hydrogens are on the same face. The two rings are fused in a way that lets the framework flex; the molecule is achiral (it has a mirror plane). In trans-decalin the hydrogens are on opposite faces and the framework is rigid — the two rings are locked. Trans-decalin is the more stable isomer because the ring strain is lower; it is also achiral, but for a different reason: it possesses a centre of symmetry.

The pair is diastereomeric, not enantiomeric. Both cis- and trans-decalin are achiral; they differ in stability, density and physical properties, and they can be separated by distillation.

KEY RESULT
Ring-junction stereochemistry is diastereomerism, not enantiomerism. Both cis- and trans-decalin are achiral; they differ in stability, shape and physical properties, and they can be separated (distillation, crystallisation).

2.24 How to attack a bicyclic drawing

1. Identify the two bridgeheads.
2. Ask whether each bridge can reach the opposite face — draw the trans attempt and see whether a bond must pass through the ring.
3. If both junctions are possible, count 2 stereoisomers (cis and trans) before looking for enantiomers.
4. Check each for σ and i; a cis junction often has a plane, a trans junction often removes it.

NOTE
Bicyclic systems also appear in the counting questions of section 6.14, where the two bridgeheads of a rigid ring must be counted as **one** stereocentre, because flipping one without the other is impossible.

2.25 The basic count

If a molecule has n independent geometrical centres and its two ends are constitutionally different, every centre doubles the number of arrangements and no two arrangements are related by symmetry:

$$N_{geo} = 2^n \text{ (dissimilar ends; n = number of qualifying centres)}$$

The formula fails in three situations, and each has its own treatment: symmetric (similar-ended) chains, centres that cannot give two isomers (C=C in a small ring), and centres that are coupled (cumulenes, where the whole chain counts as one).

2.26 Symmetric chains

For a chain whose two ends are identical, the sequence of cis/trans descriptors read from left to right is the same molecule as the sequence read from right to left. Patterns therefore merge. Counting the distinct sequences is a symmetry problem, and the result is:

$$\text{n odd: } N = 2^{(n-1)} + 2^{((n-1)/2)}$$
$$\text{n even: } N = 2^{(n-1)} + 2^{(n/2\ -\ 1)}$$
$$\text{both: } N = (2^n + 2^{\text{ceil}(n/2)}) \,/\, 2$$

n	Raw 2^n	Distinct (similar ends)	Formula check
1	2	2	$2^0 + 2^0 = 2$ (but-2-ene: cis and trans)
2	4	3	$2^1 + 2^0 = 3$
3	8	6	$2^2 + 2^1 = 6$
4	16	10	$2^3 + 2^1 = 10$
5	32	20	$2^4 + 2^2 = 20$
6	64	36	$2^5 + 2^2 = 36$

Table 2.13 — symmetric polyenes. The n = 1 row is a reminder that a single centre with identical ends is not a centre at all.

Derivation for n = 2: write the two descriptors as a pair. There are four pairs: (E,E), (E,Z), (Z,E), (Z,Z). Reading the molecule from the other end swaps the pair, so (E,Z) and (Z,E) are the same compound; the other three (E,E), (Z,Z) and (E,Z) survive. The formula gives $2^1 + 2^0 = 3$.

Derivation for n = 3 (orbit counting): eight sequences; the end-exchange operation reverses the sequence. Four sequences are palindromes (E,E,E), (E,Z,E), (Z,E,Z), (Z,Z,Z) — each is its own reverse — and the other four pair up into two orbits. Total orbits = (8 + 4)/2 = 6, which is $2^2 + 2^1 = 6$. The same argument gives $(2^n + 2^{\text{ceil}(n/2)})/2$ in every case, so you only need to remember one formula.

> TRAP
> When a question says "structural, geometrical and optical", count the skeletons first. Do not multiply 2^n by the optical count; some geometrical isomers are meso and contribute no active forms.

2.27 The traps in the count

1. **Terminal CH_2 never counts.** A double bond with two H on one carbon is not a centre.
2. **Similar ends cut the count.** Apply the symmetric formula, not 2^n.
3. **A cumulene counts as one centre, not one per double bond.**
4. **A ring junction counts as one centre if cis and trans are both possible.**
5. **Symmetrical ends merge patterns; do not use 2^n blindly.**
6. **Geometrical and optical counts are separate.** A molecule may have both; the total stereoisomer count is not simply the product of the two formulas.

2.28 Worked examples

(a) Hexa-2,4-diene, $CH_3CH{=}CH{-}CH{=}CHCH_3$. Two centres, similar ends: $2^1 + 2^0 = 3$ isomers: (E,E), (E,Z) = (Z,E), (Z,Z) — the last being the same as (Z,E).

(b) Octa-2,4,6-triene. Three centres, similar ends: $2^2 = 4$ isomers: (E,E,E), (E,E,Z), (E,Z,E), (Z,E,E) and their reverses are identical.

(c) $CH_3CH{=}CH{-}CH_2{-}CH{=}CHCH_3$. Two centres, dissimilar ends: $2^2 = 4$ isomers, because (E,Z) and (Z,E) are different compounds (the ends are not equivalent).

(d) Hexa-2,3-tetraene (an odd cumulene). One centre only: 2 isomers, the cis and trans forms of the whole cumulene.

KEY RESULT
If the question asks for the number of stereoisomers rather than geometrical isomers, add the optical possibilities. Sheet J-19 works through the combination systematically.

2.28b Counting geometrical isomers of rings and cumulenes

Rings and cumulenes are where the 2^n formula is most often misapplied, because "a centre" is not always "a double bond".

Structure	Centres	Count	Reason
1,2-dimethylcyclopropane	1 ring centre	2	cis and trans; trans is chiral
1,2-dimethylcyclohexane	1 ring centre	2	cis and trans; both achiral, trans is chiral? check: trans-1,2 has no plane → chiral pair (2), cis has a plane
1,4-dimethylcyclohexane	1 ring centre	2	cis and trans; cis has a plane
$CH_3CH{=}C{=}CHCH_3$	1 cumulene centre	2	enantiomeric pair, not cis/trans
hexa-2,4-diene	2 centres, similar ends	3	(E,E), (E,Z), (Z,Z)
hexa-2,4-diene with different ends	2 centres, different ends	4	(E,E), (E,Z), (Z,E), (Z,Z)

Table 2.14 — rings and cumulenes count as one centre each, not one per double bond.

TRAP
The trans-1,2-dimethylcyclohexane row is a trap: its two chairs are (a,a) and (e,e), but flipping does not change the up/down sense, so the trans relationship is configurational. The molecule has no plane and no centre, so it is chiral and exists as a pair — the geometrical count is still 2, but the stereoisomer count is 3.

cis and trans isomers are diastereomers: same connectivity, different 3-D shape, different physical properties. Four properties are examined again and again, and each has one controlling idea — learn the idea, not the list.

Property	Usually larger for	Controlling idea
dipole moment, μ	cis	the two bond moments add (same side) in cis and partly cancel in trans
boiling point	cis (with an important exception)	cis is more polar; dipole-dipole attraction is stronger in the liquid
melting point	trans	trans packs better laterally, so the lattice is tighter and the melting point is higher
solubility in water	cis (usually)	the cis isomer is more polar
thermodynamic stability	trans	bulky groups are further apart — less steric strain

Table 2.15 — one controlling idea per property; the exceptions follow from the same idea.

2.29 Dipole moment

A bond dipole is a vector along the bond. In cis-1,2-dichloroethene the two C–Cl bond moments point to the same side and **add**; in the trans isomer they point in opposite directions and cancel. So μ(cis) > μ(trans), and for trans-1,2-dichloroethene μ = 0. The same logic explains why cis-but-2-ene has a small dipole while trans-but-2-ene has μ = 0.

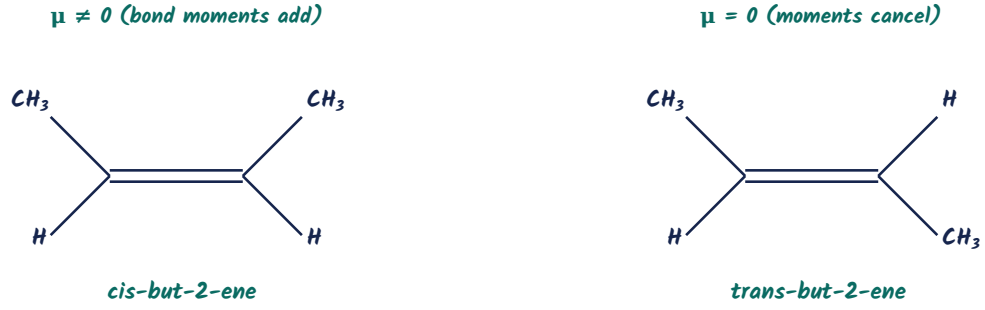


Fig. 2.4 — cis and trans but-2-ene. The C–CH_3 bond moments add in cis and cancel in trans, which is why only the cis isomer has a measurable dipole.

TRAP
Do not say "the trans isomer has zero dipole moment" as a universal rule. Trans-1,2-dichloroethene has μ = 0, but a trans isomer with two different polar groups (trans-1-chloro-1-propene, for example) has a non-zero dipole. Work the vectors.

2.30 Boiling point and solubility

Boiling requires the molecules to escape from each other, so it tracks **intermolecular attraction**: larger μ gives stronger dipole-dipole attraction, so the cis isomer normally boils higher. Water solubility tracks polarity and hydrogen bonding with water, so the cis isomer usually dissolves better as well.

The exception that JEE loves is **maleic acid versus fumaric acid**. Maleic acid is the cis diacid; its two –COOH groups are close enough to form an **intramolecular** hydrogen bond, which reduces the number of groups available for hydrogen bonding with neighbours. Fumaric acid, being trans, cannot form that internal bond, so it hydrogen-bonds **between** molecules. Fumaric acid therefore has the much higher melting point and is much less soluble in water, despite being the symmetric isomer.

KEY RESULT
Melting point follows symmetry; boiling point follows attractive forces. They usually point in opposite directions for a cis/trans pair, and an intramolecular hydrogen bond can reverse the boiling-point order.

TRAP
Do not extend the maleic/fumaric argument to boiling point. Fumaric acid sublimes and decomposes near 200 °C and does not give a clean normal boiling point, so "fumaric has the higher boiling point" is not a safe exam answer. Stick to melting point and solubility, which are measured and quoted for this pair.

2.31 Stability

Trans is normally more stable because the bulky groups are further apart. The margin is small for but-2-ene (about 4 kJ mol^{-1}) and large when the substituents are big. Two important exceptions: cis-cycloalkenes in small rings are more stable because trans is geometrically impossible, and cis-azobenzene is less stable than trans by about 50 kJ mol^{-1} because the cis form forces the phenyl rings into each other.

2.32 Separating the pair

Because cis and trans are diastereomers, their boiling points differ and they can be separated by fractional distillation; their melting points differ even more, which is why recrystallisation usually works for the solids. This is the reason a cis/trans mixture is a normal separation problem, not a resolution problem — no chiral reagent is needed.

NOTE
Numerical values in this section are standard textbook figures quoted to fix orders of magnitude (μ of cis-1,2-dichloroethene ≈ 1.9 D, μ of trans = 0; ΔH for cis → trans but-2-ene ≈ −4 kJ mol^{-1}). Do not memorise the third digit; memorise the direction and the reason.

2.33 Three worked property questions

(a) Which has the higher boiling point, cis- or trans-but-2-ene? The cis isomer has a small but real dipole (the two C–CH_3 moments do not cancel), so dipole-dipole attraction is stronger in the liquid. cis-But-2-ene boils at about 4 °C against 1 °C for trans; the difference is small because the molecule is small, with polar substituents such as Cl the gap widens.

(b) Which has the higher melting point, maleic or fumaric acid? Fumaric acid (trans) is more symmetric and packs better in the crystal, so it melts at about 287 °C (with decomposition) against about 135 °C for maleic acid. Maleic acid's intramolecular hydrogen bond also reduces its ability to form an extended lattice.

(c) Which is more soluble in water, cis- or trans-butenedioic acid? Maleic acid is more polar and its intramolecular hydrogen bond leaves one carbonyl and one hydroxyl available to water; fumaric acid's two carboxyl groups are both engaged in the intermolecular network of the crystal, so it dissolves less readily. Maleic acid is markedly more soluble.

Property	Maleic (cis)	Fumaric (trans)	Reason
melting point	≈ 135 °C	≈ 287 °C	packing / symmetry
solubility in water	high	low	polarity, H-bonding
heat of combustion	lower	higher	cis is less stable
dipole moment	larger	zero by symmetry	vector addition

Table 2.16 — the data pattern behind the rules.

TRAP
Do not transfer the maleic/fumaric boiling-point reversal to every cis/trans pair. It happens because of a specific intramolecular hydrogen bond between two -COOH groups held close. Without such a group, the cis isomer boils higher.

An optically active substance rotates the plane of vibration of plane-polarised light. Everything quantitative in optical isomerism comes from one instrument, the polarimeter, and one formula, the specific rotation.

3.1 Plane-polarised light

Ordinary light vibrates in every plane perpendicular to its direction of travel. Pass it through a Nicol prism — a calcite ($CaCO_3$) crystal cut so that only one plane survives — and the emerging light vibrates in a single plane. That is plane-polarised light (PPL). Rotate the prism and you rotate the plane.

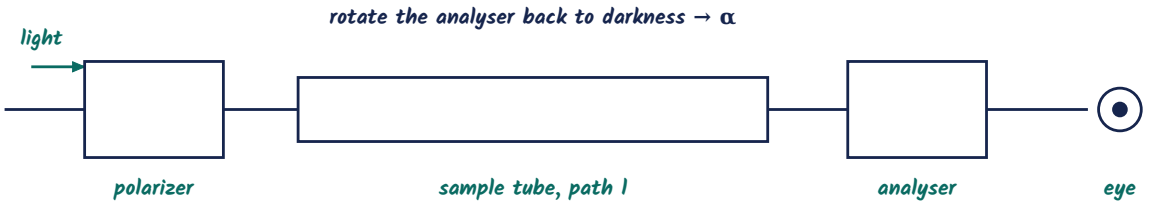


Fig. 3.1 — the polarimeter. The sample is placed between two Nicol prisms; the rotation of the analyser needed to restore darkness equals the optical rotation of the sample.

3.2 The polarimeter

1. Source gives ordinary light.
2. Polariser (first Nicol prism) converts it to PPL.
3. Sample tube of length l contains the solution of concentration c.
4. Analyser (a second Nicol prism) is rotated until the field is dark again.
5. The angle moved by the analyser is the **observed rotation** α. Clockwise, looking towards the source, is **dextrorotatory** (+); anticlockwise is **laevorotatory** (−).

An inactive sample leaves the plane unchanged, so the analyser needs no adjustment. An active sample rotates it; the two enantiomers rotate it by equal amounts in opposite directions.

3.3 Specific rotation

Observed rotation depends on how many active molecules the light meets, so it depends on the tube length, the concentration, the temperature, the solvent and the wavelength. To compare compounds we quote the **specific rotation**, defined at a stated wavelength and temperature:

$$[\alpha] = \frac{\alpha}{l \times c}$$

l in dm · c in g mL^{-1}

- α is the observed rotation in degrees, with its sign.
- l is the path length in **decimetres** (10 cm = 1 dm, so a 10 cm tube has $l = 1$).
- c is the concentration in **g mL^{-1}**, not mol L^{-1}; a value given as "5 g per 100 mL", then $c = 0.05$ g mL^{-1}.
- The standard sodium line is quoted as $[\alpha]_D$, at 589 nm.

Worked numerical. A 10 cm tube contains a solution of 5.0 g of a compound in 100 mL of solvent; the observed rotation is +6.7°. Then l = 1 dm, c = 5.00/100 = 0.0500 g mL^{-1}, and $[\alpha]$ = 6.7/(1 × 0.050) = +134°. A sample of the same compound 2.5 g per 100 mL in a 20 cm tube gives $\alpha = [\alpha] \times l \times c$ = 134 × 2 × 0.025 = +6.7° as well — the specific rotation is a property of the compound, the observed rotation is a property of the experiment.

TRAP
Units are the trap. A tube of 5 cm is l = 0.5 dm, not 5. A concentration of 10 g L^{-1} is 0.010 g mL^{-1}, not 10. And the sign must be carried through the arithmetic.

3.3b Specific rotation of a mixture, and what changes it

Because rotation is proportional to the amount of each active species, a mixture's observed rotation is the sum of the contributions. For a two-enantiomer mixture with enantiomeric excess ee, the observed specific rotation is simply ee × $[\alpha]_{pure}$ — the relation used on sheet J-14 to measure optical purity.

Worked. A sample of (S)-2-butanol shows $[\alpha]$ = +4.05° while the pure (S) enantiomer has $[\alpha]$ = +13.5°. Then ee = 4.05/13.5 = 30 %, so the sample contains 65 % (S) and 35 % (R). The observed rotation is positive, which tells you the (S) form is in excess — not that the sample is pure.

Three variables change $[\alpha]$ without changing the compound: **temperature** (quoted as $[\alpha]_D^{25}$), **solvent** (a polar solvent can hydrogen-bond to one face and shift the value), and **wavelength** (quoted as $[\alpha]_D$ for the sodium D line at 589 nm; at other wavelengths the value can differ substantially, which is the basis of optical rotatory dispersion).

Change	Effect on α	Effect on $[\alpha]$
double the tube length	doubles	unchanged
double the concentration	doubles	unchanged (dilute solutions)
change the solvent	changes	changes
change the temperature	changes	changes
change the wavelength	changes	changes
add an achiral impurity	falls	falls (concentration of active species is lower)

Table 3.1 — what is a property of the experiment and what is a property of the compound.

KEY RESULT
α belongs to the experiment; $[\alpha]$ belongs to the compound — but only when temperature, solvent and wavelength are quoted with it. An unqualified $[\alpha]$ value is incomplete.

3.4 What specific rotation does and does not tell you

The sign of [α] tells you the direction of rotation; it does not tell you the configuration. D-(+)-glyceraldehyde rotates light clockwise and has the R configuration; some R compounds are (−). Equally, the magnitude depends on wavelength and solvent, so a rotation measured in water cannot be compared with one measured in chloroform unless both are stated.

KEY RESULT
Configuration (R/S, D/L), optical rotation sign ((+)/(−)) and chirality are three different things. A molecule can be (R) and laevorotatory; a meso compound is achiral whatever you write next to it.

TRAP
A racemic mixture has α = 0 because the two rotations cancel. A meso compound has α = 0 because the molecule is achiral — internal compensation. The measured number is the same; the cause is not.

3.5 Chiral and achiral

A molecule is chiral if it cannot be superimposed on its mirror image. Your hands are the model: mirror images, never superimposable. The test is always the same: build the mirror image mentally and try to rotate it onto the original. If every atom can be brought into coincidence, the molecule is achiral and optically inactive.

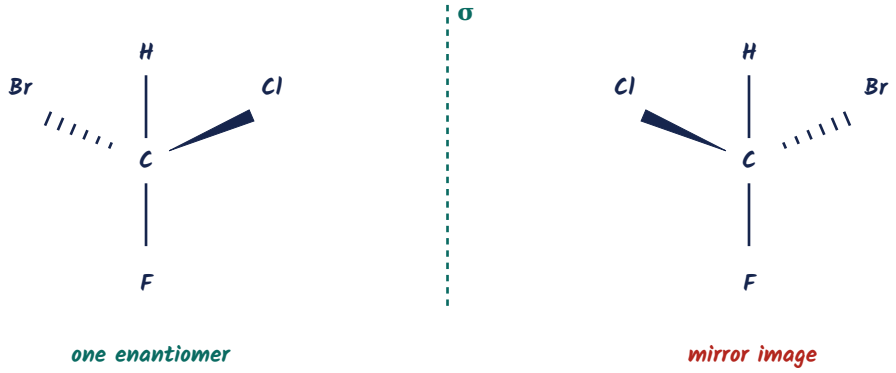


Fig. 3.2 — a carbon with four different groups and its mirror image. Three groups can always be matched; the fourth always ends up on the wrong side, which is why the pair is non-superimposable.

3.6 The chiral carbon

A tetrahedral carbon carrying four different groups is a **stereocentre** (chiral centre, asymmetric carbon). Isotopes count as different groups; H and D make a chiral carbon. Carbon atoms that are sp or sp^2 hybridised can never be stereocentres, because they do not carry four groups in a tetrahedral arrangement.

To test a candidate carbon, compare the **whole group** attached along each bond, not just the first atom. Two chains that look identical near the carbon may differ three bonds away. This is exactly the CIP comparison of section 2.8 applied to four groups.

3.7 The three rules for optical activity

Situation	Conclusion	Reason
exactly one stereocentre	always optically active	no internal plane or centre is possible
more than one, dissimilar ends	always optically active	no internal mirror plane would have to map the two ends onto each other
more than one, similar ends	may be active or meso	a symmetric conformation can supply an internal plane
no stereocentre	may still be active	axial or planar chirality (sections 3.35–3.37)

Table 3.2 — the three rules and their reasons.

KEY RESULT
One stereocentre guarantees activity. Many stereocentres do not guarantee activity, and no stereocentre does not guarantee inactivity. The only universal test is the search for symmetry elements.

3.8 Finding stereocentres in real structures

- **Open chains**: mark every carbon with four different groups, including CH(OH), $CH(NH_2)$, CH(halogen) and carbons carrying D/T.
- **Rings**: a ring carbon with two ring paths that differ (count round the ring both ways) plus H plus a substituent is a stereocentre; 1,2- and 1,4-disubstituted cyclohexanes with different substituents qualify.
- **Fused rings**: count the junction carbons whose four groups differ, but remember that a rigid junction may lock two centres together (section 6.14).
- **A carbon flanked by two stereocentres** can itself become a stereocentre, because "R" and "S" are different groups. This is the pseudoasymmetric carbon of section 6.33.
- **Nitrogen**: an amine nitrogen inverts rapidly at room temperature and is not a stereocentre; a quaternary ammonium nitrogen with four different groups is; a nitrogen in a three-membered ring inverts slowly enough to be one.

3.9 Five structures, five decisions

Structure	Stereocentre?	Why
$CH_3CH(OH)COOH$ (lactic acid)	yes, 1	C_2 carries H, OH, CH_3, COOH
$CH_3CH_2CH_2OH$	no	no carbon has four different groups
HOOC-CH(OH)-CH(OH)-COOH (tartaric)	yes, 2	C_2 and C_3, each with H, OH, COOH and the other half
$(CH_3)_2C(OH)CH_2CH_3$	no	C_2 carries two CH_3
$CH_3CH{=}CHCH_3$	no	no tetrahedral carbon with four groups

Table 3.3 — the first decision in every question: does a stereocentre exist?

3.10 Stereocentres in rings and bridged systems

In a ring, the two "ring paths" from a carbon count as two substituents. If they are different — which happens whenever the ring is unsymmetrically substituted — that carbon is a stereocentre. In a bridged bicyclic the two bridgehead carbons can be stereocentres, but because each bridgehead's configuration is tied to the other's (the ring cannot turn one without the other), the pair counts as **one** independent stereocentre in the counting formulas.

NOTE
Ring junction stereochemistry is where the optical and geometrical chapters meet: a cis junction often has a plane, and a trans junction often removes it. Sheets X-03 and X-05 take this further.

3.11 Plane of symmetry

A plane of symmetry (σ) divides the molecule into two halves that are mirror images. Its presence makes the molecule achiral, because the mirror image can be superimposed by reflecting through that plane. Look for planes in the molecular plane, perpendicular to it, and — often forgotten — through a bond or through a ring carbon.

3.12 Centre of symmetry

A centre of symmetry (i) is a point such that every atom has an identical atom on the opposite side at the same distance. Molecules with i are achiral; the mirror image is generated by inversion, which is a rotation plus a reflection in disguise. trans-But-2-ene, trans-decalin and many trans-1,4-disubstituted cyclohexanes have a centre of symmetry.

3.13 Alternating axis

An alternating axis S_n is a rotation by 360°/n followed by reflection in a plane perpendicular to the axis. S_1 is the same as a plane of symmetry, S_2 is the same as a centre of symmetry, and an S_n axis does not by itself make a molecule achiral — the only achiral cases are those that can be written with σ or i.

KEY RESULT
$S_1 = \sigma$ and $S_2 = i$. Those two identities are worth memorising because they explain why "it has an S_n axis" is not the test: after writing S_1 and S_2, the test reduces to "is there a plane or a centre?"

3.14 The activity test

1. Convert every stereocentre to (R) or (S), or convert the whole molecule to one projection.
2. Look for a plane of symmetry, including the conformation you would not normally draw.
3. Look for a centre of symmetry; in symmetric chains it often appears exactly when the two halves read opposite.
4. If neither is present, the molecule is chiral; if either is present, it is achiral.

Three facts make the test quick. First, any molecule with one stereocentre is automatically chiral, so only molecules with two or more need the test. Second, a symmetric chain with an odd number of centres has a plane through the middle centre (which is why that centre is pseudoasymmetric). Third, a symmetric chain with an even number of centres has a plane through the central bond when the two halves are opposite (R,S) — the meso case.

Molecule	Symmetry elements	Chiral?
tartaric acid (R,R)	C_2 only	yes
tartaric acid (R,S)	σ (internal), i	no (meso)
2,3-dichlorobutane (R,R)	C_2	yes
2,3-dichlorobutane (R,S)	σ	no (meso)
cis-1,2-dimethylcyclopropane	σ	no
trans-1,2-dimethylcyclopropane	C_2 only	yes
cis-1,4-dimethylcyclohexane	σ, i	no

Table 3.4 — a table worth reproducing in the exam margin before you start a counting question.

TRAP
The absence of a plane is not the same as chirality, and the presence of a C_2 axis is not a sign of chirality either — it is the absence of any improper operation (σ, i, S_n) that matters. That is why this sheet tests σ and i, and not C_n.

3.15 The four labels

Pair / species	Relationship	Physical properties	Separable by distillation?
enantiomers	non-superimposable mirror images	identical except the sign of [α]	no
diastereomers	stereoisomers that are not mirror images	different mp, bp, solubility, [α]	yes
meso compound	a single achiral molecule with stereocentres	—	it is one compound
racemic mixture	1:1 mixture of two enantiomers	same as either pure enantiomer	no (it is a mixture, not a compound)

Table 3.5 — the table examiners use for one-mark contrasts.

3.16 Meso compounds

A meso compound contains stereocentres but is achiral because an internal plane (or centre) makes the two halves cancel. The classic case is **tartaric acid**: (R,R) and (S,S) are the active pair, and (R,S) is meso. The meso form has a plane through the middle of the C_2–C_3 bond, so its mirror image is itself.

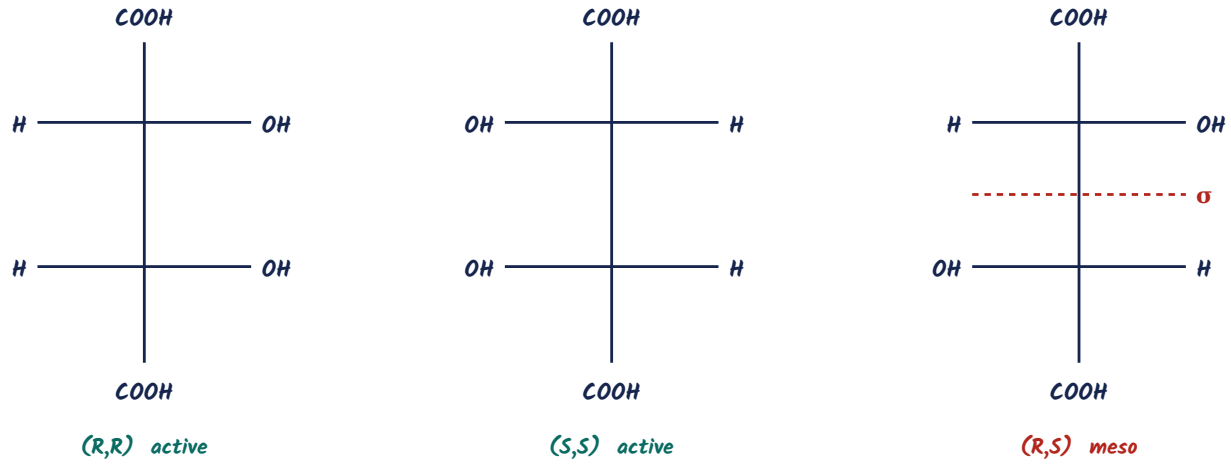


Fig. 3.3 — the three stereoisomers of tartaric acid. The (R,S) form has an internal mirror plane (dashed) and is meso; the (R,R) and (S,S) forms are the active enantiomeric pair.

Two 3-D consequences are worth remembering. First, a meso compound is a diastereomer of both active forms, so it has different physical properties and can be separated from them. Second, a meso molecule has **no optical rotation** but is not a racemic mixture — melting-point and crystallography can tell the two apart.

3.17 Racemic mixtures and compensation

A racemic mixture contains equal amounts of the two enantiomers; their rotations are equal and opposite, so the observed rotation is zero. This is **external compensation** — two different molecules cancelling. A meso compound is inactive by **internal compensation** — one molecule cancelling itself. The distinction is a standard two-mark question.

- Racemic mixtures are written (±) or (d,l).
- They form whenever a reaction creates a stereocentre in an achiral environment, because the two faces of a planar intermediate are attacked equally.
- Racemisation is the conversion of a pure enantiomer into a racemic mixture — for example by enolisation at a stereocentre, or by an S_N1 reaction through a planar carbocation.

KEY RESULT
Meso = one molecule, internally compensated. Racemic = two molecules, externally compensated. Both show α = 0; only the meso compound is achiral.

3.18 Special cases

(a) Tartaric acid. Three stereoisomers: (R,R), (S,S) active; (R,S) meso. The meso form has a plane, so it is achiral; the two active forms are enantiomers of each other and diastereomers of the meso form.

(b) 2,3-Dichlorobutane. Same pattern: (R,R) and (S,S) are the active pair, (R,S) is meso. Note that the two halogens are identical, so the ends are similar and the meso form exists; replace one Cl by Br and the ends become different and all four stereoisomers are active.

(c) 2,3-Dihydroxybutanedioic acid versus 2,3-dihydroxypentanedioic acid. The first is tartaric (similar ends, meso exists). The second has different ends, so no meso form; four stereoisomers, all active.

3.19 Meso identification, worked three ways

1. **Draw the Fischer projection flat.** If the top half is the mirror image of the bottom half across a horizontal mid-line, the compound is meso.
2. **Convert to (R)/(S).** If the descriptors are opposite at every centre (R,S or S,R) and the chain is symmetric, it is meso.
3. **Look for the plane in 3-D.** In a staggered conformation with the two halves anti, the plane through the central bond becomes visible.

All three methods must agree. The Fischer test is fastest under time pressure, but it is only valid while the projection is correctly drawn; that is why the conversion to (R)/(S) is the safe check.

3.20 Separating meso from racemic

A meso compound and a racemic mixture can both show zero rotation, but they behave differently: the meso compound is a single substance with a sharp melting point, while the racemic mixture melts over a range (and often lower than either pure enantiomer). Crystallisation of the racemic mixture may give a racemic compound (a 1:1 crystal) or a conglomerate (a mechanical mixture of the two enantiomer crystals) — the second can, in favourable cases, be separated by hand-picking or by preferential crystallisation.

TRAP
A racemic mixture does not have to be 50:50 exactly to show zero rotation; any mixture whose rotations cancel is optically inactive. The exam phrase is "optically inactive due to external compensation", not "achiral".

Enantiomers have identical physical properties in an achiral environment, so they cannot be separated by distillation, crystallisation or ordinary chromatography. Separation requires making them diastereomeric, and that means introducing a chiral influence.

3.21 The chemical method

1. React the racemic mixture with a single enantiomer of a chiral reagent (a resolving agent).
2. The two enantiomers now form two **diastereomeric** salts (or esters, or amides), which have different physical properties.
3. Separate the diastereomers by fractional crystallisation.
4. Hydrolyse each diastereomer back to the pure enantiomer and recover the resolving agent.

The classical example is the resolution of a racemic amine with (+)-tartaric acid: the two diastereomeric tartrates have different solubilities, so one crystallises first. Racemic acids are resolved with a chiral base such as brucine or quinine.

KEY RESULT
Resolution works by converting an enantiomer pair into a diastereomer pair. Diastereomers differ in physical properties; enantiomers do not. Every resolution method is a version of that sentence.

3.22 The other methods

- **Biological / enzymatic**: an enzyme is chiral and acts on only one enantiomer, so it can destroy one component (kinetic resolution) or convert it selectively.
- **Chiral chromatography**: a chiral stationary phase (cellulose derivatives, cyclodextrins) forms diastereomeric complexes with the two enantiomers, so they elute at different times.
- **Mechanical separation**: only when the racemic mixture crystallises as a conglomerate; historically the first resolution (Pasteur, tartaric acid salts, 1848).
- **Preferential crystallisation**: seeding a supersaturated racemic solution with one pure enantiomer crystal.

3.23 Optical purity

The composition of a mixture of enantiomers is expressed as **optical purity** (also called enantiomeric excess, ee):

$$\text{ee} = (\text{observed } [\alpha] \,/\, [\alpha] \text{ of the pure enantiomer}) \times 100\ \%$$

$$\text{ee} = ([\text{R}] - [\text{S}]) \,/\, ([\text{R}] + [\text{S}]) \times 100\ \%$$

A sample with $[\alpha] = +9.0°$ whose pure enantiomer has $[\alpha] = +15.0°$ has ee = 60 %, so it contains 80 % of one enantiomer and 20 % of the other. Note that ee is a ratio of rotations, not a ratio of concentrations taken directly — the two are equivalent only because rotation is proportional to concentration.

NOTE
Optical purity and enantiomeric excess are the same number when the only chiral species present are the two enantiomers. If a meso compound or an achiral impurity is present, the measured rotation no longer reports ee reliably.

3.24 Three worked numericals

(a) A solution contains 4.0 g of a compound in 50 mL, in a 20 cm tube; the observed rotation is −8.0°. Then $l = 2$ dm and $c = 0.080$ g mL^{-1}, so $[\alpha] = -8.0/(2 \times 0.080) = -50°$. The substance is laevorotatory.

(b) Pure (*R*)-butan-2-ol has $[\alpha] = +13.5°$. What rotation would a 15 cm tube containing 0.20 g mL^{-1} show? $\alpha = [\alpha] \times l \times c = 13.5 \times 1.5 \times 0.20 = +4.05°$.

(c) A 10 cm tube of a 2.0 % (w/v) solution rotates PPL by +2.4°. Then $[\alpha] = 2.4/(1 \times 0.020) = +120°$. A 2.0 % w/v solution is 2.0 g per 100 mL = 0.020 g mL^{-1}; converting correctly is the whole question.

Given	Convert to	Divide / multiply by
10 cm tube	$l = 1.00$ dm	÷ 10
20 cm tube	$l = 2.00$ dm	÷ 10
5 cm tube	$l = 0.50$ dm	÷ 10
x g per 100 mL	$c = x/100$ g mL^{-1}	÷ 100
x g L^{-1}	$c = x/1000$ g mL^{-1}	÷ 1000
x % w/v	$c = x/100$ g mL^{-1}	÷ 100

Table 3.6 — unit conversions that decide the mark.

TRAP
Blind use of the observed rotation is proportional to path length and to concentration. $c = [\alpha]\ l\ c$, which is exactly the definition rearranged. It holds for dilute solutions; at high concentration, solute-solute interactions make $[\alpha]$ non-linear in c (and $[\alpha]$ then depends slightly on the concentration at which it was measured).

3.24b Resolution in practice

Resolution is a separation, so it always costs material. Three practical points decide whether a method is usable, and they are examinable as one-line answers.

1. **The resolving agent must be enantiopure.** A partially resolved agent gives partially resolved product; the two effects multiply.
2. **The diastereomeric salts must differ enough in solubility.** If they crystallise together, fractional crystallisation fails and chromatography or a different agent is needed.
3. **The hydrolysis step must not racemise the product.** Acid or base hydrolysis of an ester or salt can epimerise a centre next to a carbonyl, which undoes the separation.

Method	Chiral influence	Scale	Exam use
diastereomeric salt	enantiopure acid or base	industrial	classic, easiest to write
enzyme (kinetic)	enzyme active site	industrial, green	"one enantiomer reacts, the other does not"
chiral HPLC/GC	chiral stationary phase	analytical to preparative	quickest to describe
mechanical / conglomerate	the crystal itself	rare	Pasteur; historical

Table 3.7 — four resolutions, one idea: make the pair diastereomeric.

KEY RESULT
Every resolution is a diastereomer separation in disguise. Write that sentence first, then give the specific method.

3.25 Fischer projection rules

1. The carbon chain is drawn **vertically**, with the most oxidised carbon at the top.
2. **Vertical bonds go back**, away from you; **horizontal bonds come forward**, towards you.
3. The intersection of the cross is a carbon; its substituents are the four bond ends.
4. A 90° rotation in the plane of the paper inverts the configuration; a 180° rotation in the plane preserves it.
5. Swapping any two groups at a centre inverts that centre; swapping twice restores it.



Fig. 3.4 — the same molecule drawn as a Fischer projection and as a wedge-dash structure. In the Fischer, the two horizontal bonds come towards you and the two vertical bonds go back.

KEY RESULT

In a Fischer projection, a 180° rotation in the plane is allowed and keeps the molecule the same; a 90° rotation is forbidden because it turns the vertical (back) bonds into horizontal (front) bonds. This one rule answers most projection questions.

3.26 Converting between projections

The safe conversion is always via the wedge-dash drawing. Build the molecule once in 3-D, then read off each projection from it. Do not try to convert Fischer directly to Newman by moving groups on the paper — that is where most errors happen.

1. Fischer → wedge-dash: put the vertical groups behind the plane and the horizontal groups in front; then rotate the whole molecule so that the main chain is zig-zag.
2. Wedge-dash → Newman: look down the bond of interest; the group on the front carbon is the centre of the front circle, the three others radiate from it.
3. Newman → sawhorse: tilt the Newman projection so that the front and back carbons are visible as points, and connect the corresponding substituents.
4. Sawhorse → Fischer: rotate about the central bond until the main chain is vertical and both ends point away from you, then flatten.

3.27 Reading a Newman projection

In a Newman projection you look down a C–C bond. The front carbon is the point where three bonds meet; the back carbon is the circle. A substituent drawn from the circle is on the back carbon, and its apparent position is the projection of the real bond. The dihedral angle is measured between a front and a back substituent, looking along the bond.

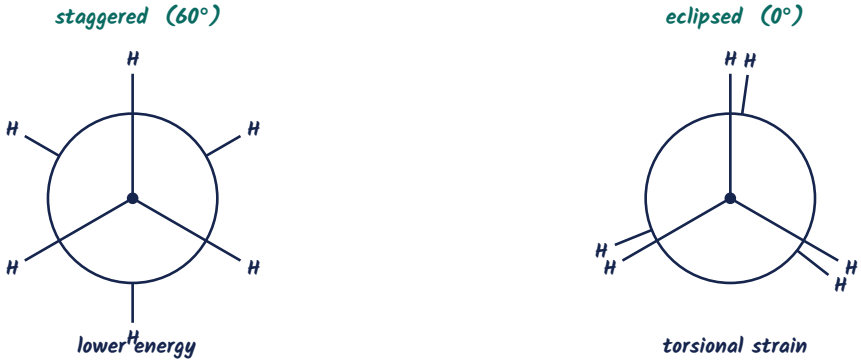


Fig. 3.5 — staggered and eclipsed Newman projections of ethane. In the eclipsed form each rear C–H hides behind a front C–H; that is the source of torsional strain.

3.28 Two worked conversions

(a) (R)-lactic acid as a Fischer. Put COOH at the top and CH_3 at the bottom; H and OH go on the horizontal arms. With OH on the right and H on the left, the priorities are OH (1) > COOH (2) > CH_3 (3) > H (4). H is on a horizontal arm, so it points towards you; read the other three in the direction 1 → 2 → 3 and reverse the answer. 1 → 2 → 3 runs clockwise → S, reversed → R.

(b) The same molecule as a Newman projection. Looking down C_2–C_3 with COOH at the top of the front carbon, the OH and H are arranged so that OH is on the right and H on the left; the back carbon carries three H. Rotating the front carbon to bring H to the back position reproduces the Fischer arrangement, which is the check that the conversion is right.

> TRAP
> Never rotate a Fischer projection by 90° to make it fit the page. A 90° rotation converts every wedge into a dash and gives the enantiomer. Rotate by 180°, or redraw from the 3-D structure.

3.29 The four steps

1. Rank the four groups on the stereocentre by CIP priority (section 2.8).
2. Look along the bond from the stereocentre to the **lowest-priority group** (usually H), so that H points away from you.
3. Read the direction 1 → 2 → 3: clockwise is **R** (rectus), anticlockwise is **S** (sinister).
4. If the lowest-priority group points towards you, reverse the answer.

KEY RESULT
R/S is a property of the whole configuration, not of a drawing. Any rotation, any projection, any conformation of the same molecule gives the same descriptor — that is exactly why it is useful.

3.30 Worked examples

(a) 2-Butanol, $CH_3CH(OH)CH_2CH_3$. Priorities: OH (1) > CH_2CH_3 (2) > CH_3 (3) > H (4). With H pointing away and the sequence 1 → 2 → 3 clockwise, the configuration is (R).

(b) Glyceraldehyde, $HOCH_2CH(OH)CHO$. Priorities on C_2: OH (1) > CHO (2) > CH_2OH (3) > H (4) — the aldehyde carbon counts as (O, O, H) and beats CH_2OH, which is (O, H, H). D-Glyceraldehyde is (R).

(c) A Fischer with H horizontal. If H is on a horizontal arm it points towards you, so read 1 → 2 → 3 and reverse. A clockwise 1 → 2 → 3 with H horizontal is (S).

(d) A molecule with two centres. Assign each centre separately; the molecule's name lists both, e.g. (2R,3S)-2,3-dichloropentane. Do not try to assign "the molecule" a single descriptor.

3.31 R/S for special centres

Centre	How to rank	Note
carbon with D/T	D > H; T > D	isotopes are different groups
nitrogen (quaternary)	four groups, no lone pair	a stereocentre if all four differ
nitrogen (amine)	three groups + lone pair	inverts rapidly; not a stereocentre unless in a small ring
sulfur (sulfoxide)	lone pair is the lowest ligand	sulfoxides are configurationally stable
phosphorus (phosphine)	lone pair lowest	inverts slowly; can be resolved
allene end	axial, not a centre	use a/d or the axial R/S convention of section 6.8

Table 3.8 — centres beyond carbon, and why some of them are not centres.

TRAP
The lone pair is always the lowest-priority "group". A nitrogen with three different groups plus a lone pair has a configuration, but in an ordinary amine it inverts at about 10^{10} s^{-1} at room temperature, so it is not a resolvable stereocentre.

3.32 Relative configuration

Before R/S existed, configurations were related to a standard: **D-(+)-glyceraldehyde**. A compound whose highest-priority stereocentre has the same arrangement as D-glyceraldehyde is given the D prefix; the opposite is L. This is relative configuration: it names the family, not the absolute arrangement.

The convention survives because nature uses it. Amino acids are almost all L, sugars are mostly D, and the phrases "D-sugar" and "L-amino acid" carry biochemical meaning that R/S does not.

Prefix	What it fixes	What it does not fix
D / L	the arrangement at the highest-priority centre relative to glyceraldehyde	the sign of rotation
(+) / (−)	the direction the compound rotates PPL	the configuration
R / S	the absolute configuration at each centre	the sign of rotation

Table 3.9 — three independent labels that are constantly confused.

KEY RESULT
D and L do not mean (+)- and (−)-. D-Glyceraldehyde happens to be (+), but D-lactic acid is (−). Any statement that equates D/L with +/− is wrong.

3.33 threo and erythro

For a chain with two adjacent stereocentres carrying the same groups, the older names describe the relationship between the two substituents in the Fischer projection. If the two identical groups are on the **same side**, the arrangement is **erythro**; if they are on opposite sides, it is **threo**. The names come from the sugars erythrose and threose.

- erythro: the two like groups are on the same side of the Fischer projection.
- threo: the two like groups are on opposite sides.
- The names are **relative** — they describe one diastereomer relative to the other, and each of them may be chiral or meso depending on the ends.
- For 2,3-dichlorobutane the erythro form is the meso one (a plane), and the threo form is the active pair.

3.34 Why the older names survive

Three reasons. First, the names describe a **relationship** that R/S does not state directly: whether two substituents are on the same face. Second, they are shorter to say for long chains. Third, they are used in the carbohydrate and amino-acid literature, where the relative arrangement is what matters chemically (for example, in the erythro/threo selectivity of aldol reactions).

The exam-safe approach is to give both: "the threo pair, i.e. (2R,3R) and (2S,3S)". If a question uses threo/erythro without defining the reference groups, state your definition in one line before answering.

NOTE
threo/erythro are not the same as syn/anti. syn/anti describe a double bond or a ring face; threo/erythro describe two adjacent tetrahedral centres. Confusing them is a common way to lose a mark on a straightforward question.

TRAP
Do not assume that erythro always means meso. Erythro means "like groups on the same side"; whether that gives a plane depends on the ends. With different ends, the erythro form is chiral as well.

Every counterexample to "chiral means a chiral carbon" is built the same way: a **locked axis or plane**, two unsymmetrical ends, and no internal mirror plane. Once you have seen the pattern, the compounds stop looking exceptional.

3.35 Allenes

An allene C=C=C has two perpendicular end planes. If one end carries a, b (a ≠ b) and the other carries x, y (x ≠ y), the molecule has no plane and no centre, and the two arrangements are non-superimposable mirror images. The chirality belongs to the **axis**, not to any atom. The descriptors used for such compounds are a/d or the axial R/S convention; JEE asks only that you recognise chirality and count the two enantiomers.

KEY RESULT

Allene rule: both ends unsymmetrical → chiral (one pair). Either end symmetric → achiral. This is the same odd-even argument as section 2.18, used for the optical chapter.

3.36 Spiro compounds

A spiro carbon joins two rings in perpendicular planes. If each ring is unsymmetrically substituted (different groups on the two sides of the spiro atom), the molecule has axial chirality; if either ring has a plane through the spiro atom, the molecule is achiral. The test is therefore per ring, not per molecule.

The classic exam case is spiro[3.3]heptane with two different substituents on each ring — substituted cyclobutanes — which exists as a pair of enantiomers. The spiro carbon itself is not a stereocentre: it carries two identical ring paths on each side only if the rings are symmetric, but even when they are not, the chirality is axial, not central.

3.37 Substituted biphenyls

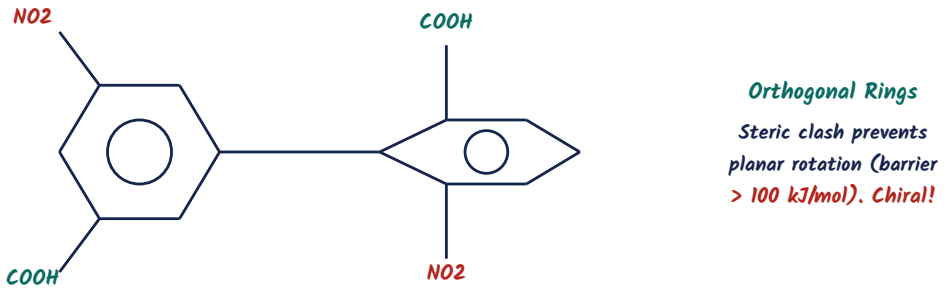


Fig. 3.2b — Atropisomerism in ortho-substituted biphenyls: steric hindrance prevents planar rotation.

Two benzene rings joined by a single bond can rotate about it. Bulky substituents in the ortho positions prevent that rotation, and the rings lock at an angle to each other. With four different ortho substituents the two locked arrangements are non-superimposable mirror images — **atropisomers** (axial chirality from restricted rotation).

The barrier for 2,2′-disubstituted biphenyls is typically 80–160 kJ mol^{-1}, enough to isolate the enantiomers at room temperature. Key exam examples: 6,6′-dinitro-2,2′-diphenic acid and 2,2′,6,6′-tetramethylbiphenyl-3,3′-dicarboxylic acid (Mills–Nixon). If all four ortho positions carry the same group, the molecule has a plane and is achiral.

Structure	Chiral?	Reason
biphenyl	no	free rotation; planar average
2,2′-dimethylbiphenyl	no	two identical halves → plane
2,2′-dimethyl-6,6′-dinitrobiphenyl	yes	four different ortho groups; rotation locked
2,2′,6,6′-tetramethylbiphenyl	no	symmetrical; plane perpendicular to the axis
1,1′-binaphthyl-2,2′-diol (BINOL)	yes	two naphthyl halves, no mirror plane

Table 3.10 — the substituent pattern decides, not the ring size.

3.38 Helicenes and propellers

A helicene is a chain of ortho-fused benzene rings that cannot lie flat; it coils like a spring, and the two coils are non-superimposable mirror images. Propeller-shaped molecules (three or four rings twisted the same way about a central bond) are chiral for the same reason. Both are chiral because the whole shape is handed, not because of any one atom.

3.39 Metal complexes

An octahedral complex with three bidentate ligands, $[M(en)_3]^{n+}$, is chiral: the three chelate rings can wind clockwise or anticlockwise, giving the Λ and Δ enantiomers. The metal carries no "four different groups" in the organic sense. This is the inorganic face of the same idea and is examined in coordination chemistry.

NOTE
Chirality is a property of the **arrangement**, not of an atom. Any statement that "chiral molecules must contain a chiral carbon" is wrong, and the standard counterexamples are allene, the ortho-substituted biphenyl and the Λ/Δ complex.

3.39b The four kinds of chirality

It helps to sort every chiral molecule into one of four families. The families differ in where the handedness lives, and each has its own quick test.

Kind	Where the handedness lives	Test	Example
central	one atom with four different groups	find the stereocentre	lactic acid
axial	a locked axis with unsymmetrical ends	both ends differ, rotation blocked	allene, BINOL
planar	a plane that is not a mirror plane	substituents arranged above/below a plane	trans-cyclooctene, cyclophanes
helical	a screw-shaped framework	the coil cannot be flattened	helicene, some polymers

Table 3.11 — the four families; every chiral molecule belongs to one of them.

Planar chirality deserves one line of explanation because it is easy to miss. A molecule such as a cyclophane can have a plane that contains the ring but is not a mirror plane, because the substituents above and below it are different. The molecule is then chiral even though no single atom carries four different groups and no axis is locked. Trans-cyclooctene is the smallest examinable example: the double bond forces the ring above and below the alkene plane, and the two enantiomers are configurationally stable at room temperature.

KEY RESULT
Ask "where is the handedness?" rather than "where is the chiral carbon?". Central, axial, planar and helical chirality cover every case JEE sets.

3.40 The counting table

Let n be the number of **independent** stereocentres. If the two ends of the molecule are constitutionally different, no meso form is possible. If the ends are the same, half of the 2^n strings are the same molecule written from the other end, and the palindromic ones are meso.

Case	Optically active	Meso (inactive)	Total stereoisomers
different ends (any n)	2^n	0	2^n
similar ends, n even	$2^{(n-1)}$	$2^{(n/2\ -\ 1)}$	$2^{(n-1)} + 2^{(n/2\ -\ 1)}$
similar ends, n odd	$2^{(n-1)} - 2^{((n-1)/2)}$	$2^{((n-1)/2)}$	$2^{(n-1)}$

Table 3.12 — the counting table. Burnside's lemma (sheet X-07) gives the same numbers: $N = (2^n + 2^{\text{ceil}(n/2)})/2$.

The table is easier to remember if you think of the **total** first. With similar ends the total is $2^{(n-1)} + 2^{(n/2-1)}$ for even n and $2^{(n-1)}$ for odd n; the meso members are the palindromic strings, and the rest come in enantiomeric pairs. Check with tartaric acid, n = 2: total 2 + 1 = 3 — the active pair (R,R)/(S,S) and one meso (R,S).

n	Ends	Active	Meso	Total	Example
1	—	2	0	2	lactic acid
2	different	4	0	4	2-chloro-3-bromobutane
2	similar	2	1	3	tartaric acid
3	different	8	0	8	2,3,4-trihydroxypentanedioic acid
3	similar	2	2	4	trihydroxyglutaric acid
4	similar	8	2	10	four-centre symmetric diol

Table 3.13 — values to check your formula against.

3.41 Worked example: trihydroxyglutaric acid

HOOC-CH(OH)-CH(OH)-CH(OH)-COOH has three stereocentres and similar ends. n = 3 odd: active = $2^2 - 2^1 = 2$, meso = $2^1 = 2$, total = 4. The single active pair is (R,R,R)/(S,S,S). The two meso forms both have an (R) end and an (S) end with a middle carbon that is **pseudoasymmetric** — its two halves are (R) and (S), which count as different groups — so (R,r,S) and (R,s,S) are diastereomers and are counted separately. That is why the meso column reads 2 and not 1, and it is the one place where the odd-n row differs from the even-n row in kind, not just in arithmetic.

Note the contrast with geometrical isomerism of a symmetric triene: octa-2,4,6-triene has $2^{(3-1)} + 2^{((3-1)/2)} = 6$ geometrical isomers, while a three-centre chain with pseudoasymmetry has 4 optical isomers. The two counts use different equivalence rules because a cis/trans descriptor is a two-state label on a bond, whereas R/S is tied to an atom whose neighbours may themselves be chiral. Do not mix the two formulas.

3.42 A four-centre symmetric chain

HOOC-CH(OH)-CH(OH)-CH(OH)-CH(OH)-COOH has n = 4 and similar ends. Active = $2^3 = 8$, meso = $2^1 = 2$, total = 10. Which strings are meso? A compound is achiral only if its mirror image can be written as the same molecule from the other end, that is if s = invert(reverse(s)). For n = 4 that forces each half to be the mirror of the other: (R,S,R,S) ≡ (S,R,S,R) is one meso compound and (R,R,S,S) ≡ (S,S,R,R) is the other, each with a plane through the middle of the chain. Every remaining string is paired with a non-identical mirror image, which is where the 8 active forms come from.

KEY RESULT
Two routes must agree. Closed formulas (the table) and orbit counting (Burnside) give the same number for every n. If they disagree, your n is wrong or the ends are not really similar.

3.43 Combined geometrical and optical counting

1. Count the geometrical isomers first, treating each as a separate skeleton.
2. For each geometrical isomer, count its stereocentres and apply the table.
3. Add up. Some geometrical isomers may be meso (a cis ring with a plane) and contribute no active forms.

Worked: 2,4-hexadien-1-ol type chain. Two C=C centres give three geometrical isomers (similar ends). Each has no stereocentre, so the total is 3 stereoisomers, all achiral. **Worked: 2,4-dichlorohexa-2,4-diene.** Two C=C and two stereocentres, with each C=C giving a geometrical choice — the enumeration is skeleton by skeleton, not a single formula.

TRAP
The formulas count **stereocentres**, not geometrical centres. A molecule with a C=C and two stereocentres has 3 geometrical isomers and, for each, up to four optical possibilities; do not multiply n's.

KEY RESULT
Never multiply "geometrical × optical" blindly. Enumerate the skeletons first, then test each for symmetry; the formulas are shortcuts for symmetric chains, not a substitute for drawing.

3.43b Four worked counts you can check against the table

1. **2,3-Dichlorobutane**: two centres, similar ends (CH_3 at both ends). n = 2 even → active 2, meso 1, total 3.
2. **2-Chloro-3-bromobutane**: two centres, different ends. n = 2, different ends → active 4, meso 0, total 4.
3. **2,3,4-Tribromopentane**: three centres, similar ends. n = 3 odd → active $2^2 - 2^1 = 2$, meso 2, total 4.
4. **2,3,4-Tribromohexane**: three centres, different ends (CH_3 and C_2H_5). n = 3 → active 8, meso 0, total 8.

The pattern to notice is how much the ends matter: the same three-centre skeleton gives 4 stereoisomers with similar ends and 8 with different ends. That is the single most examined consequence of the counting table, and it is why the first question to ask is always "are the two ends constitutionally identical?".

Molecule	n	Ends	Active	Meso	Total
2,3-dichlorobutane	2	similar	2	1	3
2-chloro-3-bromobutane	2	different	4	0	4
2,3,4-tribromopentane	3	similar	2	2	4
2,3,4-tribromohexane	3	different	8	0	8
tartaric acid	2	similar	2	1	3
trihydroxyglutaric acid	3	similar	2	2	4

Table 3.14 — the same skeleton with different ends; check your formula against every row.

TRAP
"Different ends" means constitutionally different, not merely drawn differently. 2,3,4-tribromopentane has a CH_3 at each end and is similar; 2,3,4-tribromohexane has CH_3 at one end and C_2H_5 at the other and is different. A CH_2 inserted at one end changes the answer from 4 to 8.

A large family of objective questions gives two or four drawings and asks for their relationship. There is one reliable procedure.

3.44 The four-step procedure

1. **Molecular formula and connectivity.** If they differ, the compounds are structural isomers or unrelated — stop.
2. **Convert both to the same projection** (Fischer is usually fastest for chains, Newman for conformations).
3. **Try the allowed operations** on one structure: 180° rotation of the whole Fischer, two swaps at a centre, rotation about a single bond. If it becomes the other, they are **identical**.
4. **Compare centre by centre.** If every centre is inverted and the drawings are mirror images, they are **enantiomers**. If some centres match and some are inverted, they are **diastereomers**.

Observation after normalising	Relationship
superimposable by rotation / even swaps	identical (same compound)
mirror images, no rotation superimposes them	enantiomers
some centres same, some inverted	diastereomers
same configuration, different rotation about a single bond	conformers (identical compound)
different connectivity	structural isomers

Table 3.15 — the decision table for relation questions.

3.45 Worked comparisons

(a) Two Fischer projections that differ by a 180° rotation. They are the same molecule. A 180° rotation in the plane is allowed and changes nothing; a 90° rotation would give the enantiomer.

(b) Two Fischer projections with one centre swapped. Diastereomers. Swapping two groups at a single centre inverts that centre only, which is the definition of a diastereomeric change.

(c) (R)-lactic acid and (S)-lactic acid. Enantiomers: one centre, every descriptor inverted, mirror images.

(d) (2R,3R)- and (2R,3S)-tartaric acid. Diastereomers: one centre the same, one inverted. The second is the meso form, so the pair is also "active versus inactive", which is worth saying in a descriptive answer.

(e) A wedge-dash drawing and its mirror image, both rotated. Rotate one in the plane and compare; if the wedges now coincide, they are identical. This is where most candidates lose a mark by assuming every mirror image is an enantiomer.

3.46 A fast symmetry check

Before doing any R/S assignment, look for a horizontal mirror line in the Fischer projection. If the top half is the mirror of the bottom half, the molecule is meso and its mirror image is itself — which immediately answers "identical or enantiomer?" without any assignment. Use this as a five-second filter, then confirm with R/S.

TRAP
"Not superimposable" is not the same as "enantiomers". Two diastereomers are also non-superimposable. The definition of enantiomers needs both facts: mirror images **and** non-superimposable.

KEY RESULT
In a relation question, always state the evidence: "same formula, same connectivity, all centres inverted, mirror images → enantiomers". The reason earns the mark; the label alone may not.

Configuration is the arrangement that can be changed only by breaking bonds. Conformation is the arrangement that changes by rotating about single bonds. Everything in sections 4 and 5 depends on keeping those two words apart.

4.1 Why rotation has a barrier

Even a single bond resists rotation a little. When two bonds on adjacent carbons line up, their electron pairs repel — **torsional strain** — and the energy rises. The barrier is the height of that rise. Ethane's barrier is about 12.5 kJ mol^{-1} (textbook value; modern measurements give ≈ 12.1); butane's highest barrier, the methyl–methyl eclipse, is about 19–25 kJ mol^{-1}.

Bond	Barrier ≈	Consequence at 25 °C
C–C in ethane	12.5 kJ mol^{-1}	free rotation; conformers not isolable
C–C in butane	19–25 kJ mol^{-1}	free rotation; anti favoured
C–N in amides	≈ 75 kJ mol^{-1}	slow rotation; cis/trans amides separable
C=C π bond	250–270 kJ mol^{-1}	no rotation; geometrical isomers isolable
biphenyl, 2,2′-disubstituted	80–160 kJ mol^{-1}	atropisomers isolable

Table 4.1 — the barrier decides whether an arrangement is a conformer or a configuration.

KEY RESULT
A conformer becomes a configuration when its barrier is high enough to isolate it at room temperature (roughly above 80 kJ mol^{-1}). The same geometric idea — a locked arrangement — can therefore be conformational in one molecule and configurational in another.

4.2 What raises the barrier

- **Partial double-bond character**: amides, esters, enamines, and the C–N bond of a thioamide.
- **Steric bulk on both sides**: ortho-substituted biphenyls, tri-tert-butylmethane.
- **Ring constraints**: a bond inside a small ring cannot rotate without breaking the ring.

4.3 Representing conformations

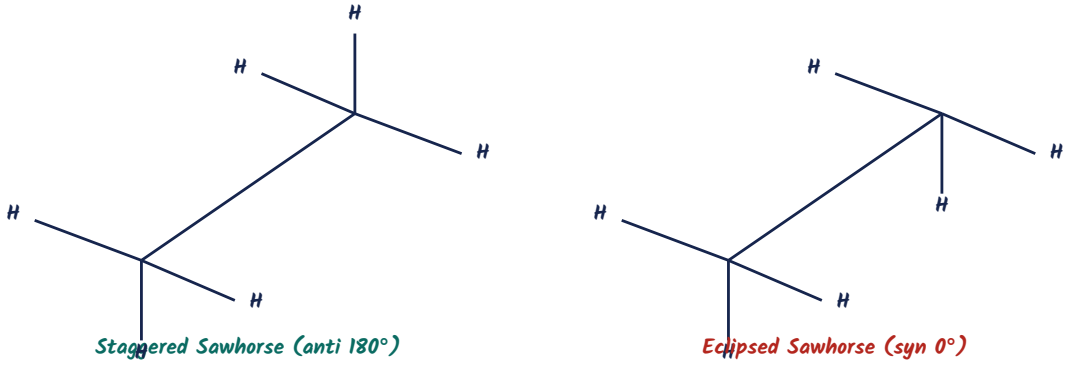


Fig. 4.0 — Staggered vs Eclipsed conformations shown in Sawhorse projection.

Three representations are used, and each has a rule that must be respected. In a **Newman** projection you look down the bond: front carbon as a point, back carbon as a circle. In a **sawhorse** drawing you look from the side: both carbons visible, bonds drawn at an angle. In a **wedge-dash** drawing the chain is in the plane and the substituents are above/below it.
In a Newman projection, the dihedral angle θ is the angle between a chosen front substituent and a chosen back substituent, measured looking down the bond. θ = 0° is eclipsed, θ = 60° is staggered. The energy profile is a periodic function of θ with three maxima and three minima per 360° for ethane.

NOTE
When you convert a sawhorse to a Newman, keep the same viewing direction. Rotating the whole molecule in space is allowed; rotating only the back carbon changes the conformation you are describing.

4.4 Ethane

Eclipsed ethane has three H–H eclipses and is about 12.5 kJ mol⁻¹ above staggered. Staggered ethane has three equivalent conformations related by 120° rotation and is the minimum. The barrier is small because the only interaction is between C–H σ bonds; there is no steric bulk.

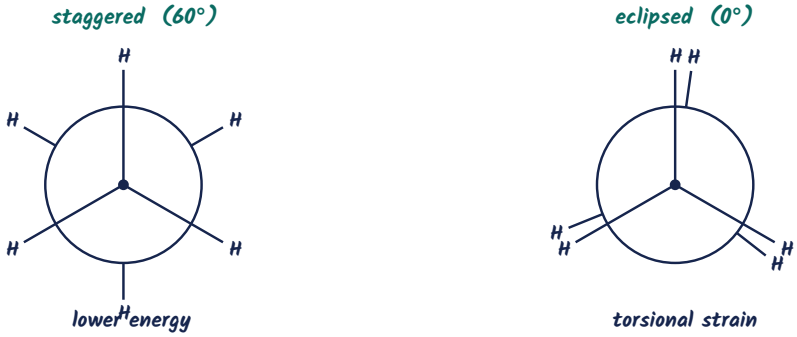


Fig. 4.1 — staggered and eclipsed ethane in Newman projection. In the eclipsed form each rear C–H hides behind a front C–H; that is the source of torsional strain.

4.5 Butane

Rotation about C_2–C_3 in butane has three distinct staggered conformers: **anti** (θ = 180°, methyls opposite) and two **gauche** forms (θ = 60° and 300°, methyls 60° apart). Anti is the global minimum; each gauche is about 3.8 kJ mol⁻¹ higher because of the methyl–methyl steric interaction. The maxima are the eclipsed forms, with the CH_3/CH_3 eclipse highest.

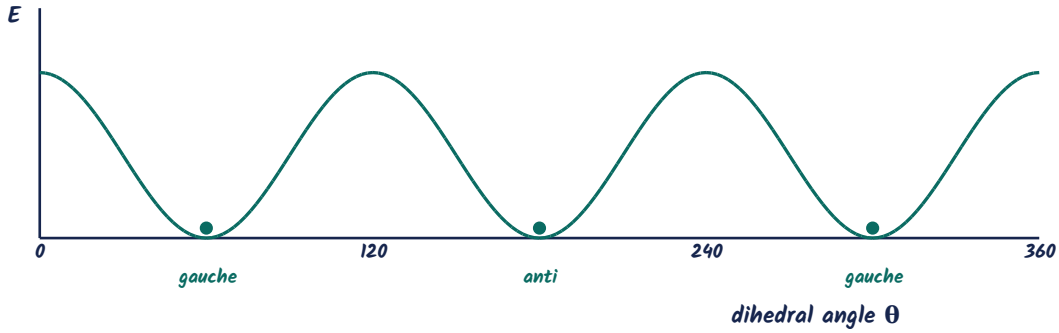


Fig. 4.2 — the potential energy of butane as the C_2–C_3 bond rotates. Minima at 60° (gauche), 180° (anti) and 300° (gauche); maxima at 0° (syn) and 120°/240° (CH_3/H eclipses).

Conformer (θ)	Relative energy	Why
anti (180°)	0	methyls farthest apart
gauche (60°, 300°)	≈ 3.8 kJ mol⁻¹	one CH_3/CH_3 gauche interaction
H/CH_3 eclipsed (120°, 240°)	≈ 16 kJ mol⁻¹	one methyl eclipse + torsional strain
CH_3/CH_3 eclipsed (0°)	≈ 19–25 kJ mol⁻¹	methyl eclipse + full torsional strain

Table 4.2 — butane's energy ladder. Populations follow Boltzmann (sheet X-04).

At room temperature the anti form is favoured but not exclusively populated: with ΔE ≈ 3.8 kJ mol⁻¹, the Boltzmann ratio anti : gauche ≈ 4.6 : 1 per gauche form, so about 70 % anti and 30 % gauche in total. The rotation is fast, so the NMR spectrum shows one averaged methyl signal, not separate signals for each conformer.

4.6 Are conformers ever chiral?

Yes — and this is a favourite JEE Advanced point. The gauche conformer of butane has no plane and no centre, so it is chiral; the two gauche forms are enantiomers. But they interconvert rapidly by rotation through the anti form, so butane itself is achiral (a conformational average) and cannot be resolved. The molecule is achiral; the conformer is chiral. The lesson: chirality can be a property of a conformation, but optical activity requires the chiral species to be long-lived.

KEY RESULT
A molecule is achiral if it has a conformation with a plane — even if other conformations are chiral — because the interconversion is fast. Optical activity needs a configurational lock, not a conformational one.

4.7 Barrier data worth remembering

Quantity	Value	Use
ethane barrier	≈ 12.5 kJ mol^{-1}	benchmark for torsional strain
butane anti → gauche	≈ 3.8 kJ mol^{-1}	benchmark for a gauche CH_3/CH_3 interaction
cyclohexane chair → twist-boat	≈ 23 kJ mol^{-1}	chair is strongly favoured
cyclohexane chair → boat	≈ 29 kJ mol^{-1}	boat is not a minimum
methylcyclohexane axial → equatorial	≈ 7.3 kJ mol^{-1}	A-value of CH_3
cyclohexane ring flip	≈ 45 kJ mol^{-1}	fast at room temperature

Table 4.3 — values to order, not to measure.

4.7b Populations and the shape of the curve

The energy profile is only half the story; the population of each conformer follows the Boltzmann distribution, $N_i/N_j = \exp(-\Delta E/RT)$. At 298 K, $RT \approx 2.48$ kJ mol^{-1}, so a 3.8 kJ mol^{-1} gauche penalty gives a ratio of $\exp(-3.8/2.48) \approx 0.22$, i.e. each gauche form is about 22 % as populated as anti.

Worked for butane: anti 1.0, two gauche forms 0.22 each, total 1.44, so anti ≈ 69 % and the two gauche forms ≈ 31 % combined. Rotamers interconvert quickly, so a physical measurement sees the weighted average — which is why the observed dipole moment of a flexible molecule rarely matches any single conformer.

ΔE / kJ mol^{-1}	Ratio at 298 K	Population of the higher form
1	0.67	40 %
2	0.45	31 %
3.8	0.22	18 %
7.3	0.052	5 %
20	3×10^{-4}	0.03 %

Table 4.4 — the Boltzmann ladder; 20 kJ mol^{-1} is effectively a lock, 1 kJ mol^{-1} is a mixture.

KEY RESULT
Order the conformers by energy, then convert to populations with $\exp(-\Delta E/RT)$ at 298 K ($RT \approx 2.5$ kJ mol^{-1}). A 5.7 kJ mol^{-1} gap is a 90:10 mixture.

4.8 Ring strain

A ring is strained when its bond angles depart from 109.5° or when its hydrogens are forced into eclipsed arrangements. Cyclopropane has angle strain (60° versus 109.5°) and torsional strain (all C–H eclipsed); cyclobutane is puckered to relieve some of it; cyclopentane is a floppy envelope; cyclohexane can adopt the strain-free chair.

Ring	Strain ≈	Shape	Note
3	115 kJ mol^{-1}	planar triangle	very high; ring-opening reactions
4	110 kJ mol^{-1}	puckered	still high
5	26 kJ mol^{-1}	envelope	small
6	0	chair	strain-free
7–12	20–50 kJ mol^{-1}	various	medium-ring strain

Table 4.5 — the strain ladder; cyclohexane is the special case, not the rule.

4.9 The chair

The chair has all bond angles at 109.5° and all adjacent C–H bonds staggered, so it has no angle or torsional strain. Each carbon carries one **axial** bond (vertical, parallel to the C_3 axis) and one **equatorial** bond (roughly in the ring plane, pointing outwards). Axial bonds alternate up/down around the ring; equatorial bonds do too, in the opposite sense.

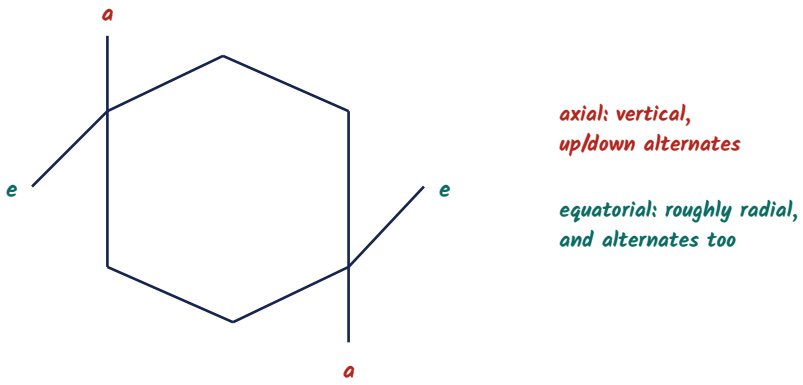


Fig. 4.3 — axial and equatorial bonds on a cyclohexane chair. At each carbon one bond is vertical (axial) and one is roughly radial (equatorial); the up/down sense alternates round the ring.

4.10 Ring flipping

A chair flips to the other chair by passing through a half-chair and a twist-boat. In the flip, every axial bond becomes equatorial and every equatorial becomes axial, but the **up/down sense is preserved**: an "up" substituent stays up. The barrier is about 45 kJ mol^{-1}, so the flip is fast at room temperature (about 10^5 s^{-1}) and the two chairs are conformers, not isomers.

KEY RESULT

Flip: a ⇄ e, e ⇄ a, up stays up, down stays down. This one sentence answers most ring-flip questions, including why a 1,2-disubstituted cyclohexane has the same cis/trans label in both chairs.

4.11 Worked chair problems

(a) Which chair of methylcyclohexane is more stable? The one with CH_3 equatorial. The axial conformer has two 1,3-diaxial H···CH_3 interactions, worth about 7.3 kJ mol^{-1} in total, so the equatorial form dominates (≈ 95 % at 25 °C).

(b) Is cis-1,2-dimethylcyclohexane chiral? No. Its more stable chair has one CH_3 axial and one equatorial, and the ring has a plane of symmetry; the molecule is meso. The trans isomer (e,e or a,a) is chiral and exists as a pair.

(c) Which is more stable, cis- or trans-1,4-dimethylcyclohexane? Trans, because it can place both methyls equatorial (e,e). Cis-1,4 must be a,e.

4.12 The boat, the twist-boat and the actual path

The boat has two flagpole hydrogens pointing at each other, which costs about 29 kJ mol^{-1} relative to the chair; the twist-boat relieves that by twisting, but is still about 23 kJ mol^{-1} above the chair and is not a minimum on the true path. The real flip goes chair → half-chair (≈ 45 kJ mol^{-1}, the transition state) → twist-boat → half-chair → chair.

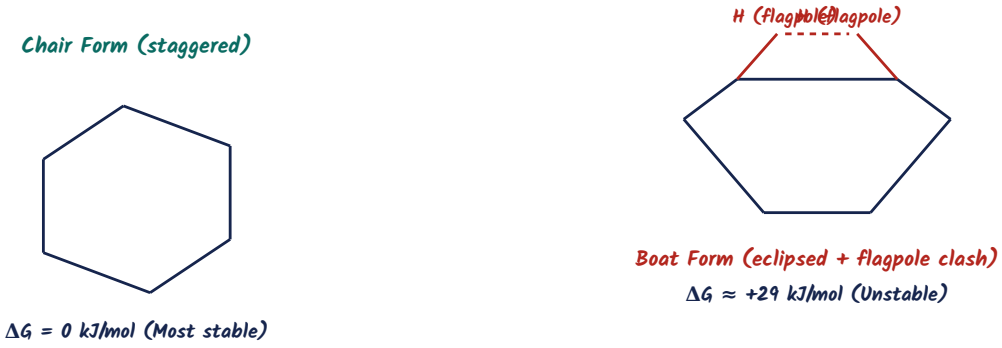


Fig. 4.4 — Chair vs Boat cyclohexane: Flagpole interactions and steric torsional strain.

NOTE
The boat is often drawn as if it were an intermediate. It is not on the minimum-energy path; the twist-boat is the shallow minimum between the two chairs, and the half-chair is the barrier.

4.13 What NMR sees

At room temperature the ring flip is fast on the NMR timescale, so axial and equatorial hydrogens average and a monosubstituted cyclohexane shows one set of signals. Cool the sample and the flip slows; the axial and equatorial protons then separate, which is the classic variable-temperature NMR experiment for cyclohexane conformational analysis.

4.14 The placement rules

1. Draw the chair and mark each carbon up or down.
2. Draw both chairs (after a flip) and place the substituent axial in one and equatorial in the other.
3. Choose the chair with the larger group equatorial.
4. For two substituents, check whether the required up/down pattern gives a,e or e,e — that decides stability.

Relationship	cis	trans	More stable
1,2	a,e (one up, one down on adjacent carbons is trans — see note)	e,e or a,a	trans (e,e)
1,3	e,e or a,a	a,e	cis (e,e)
1,4	a,e	e,e or a,a	trans (e,e)

Table 4.6 — for identical substituents. Check the up/down pattern on a drawing before quoting this table.

The patterns are easy to remember as a sequence: 1,2-cis is a,e; 1,3-cis is e,e; 1,4-cis is a,e. Trans is the opposite in each case. The reason is the alternation of the up/down sense around the ring: going one carbon changes the sense, going two carbons keeps it.

4.15 Why the pattern alternates

Label the carbons C_1 to C_6. On C_1 axial is up and equatorial is down. On C_2 axial is down, equatorial up; on C_3 axial is up, and so on. A cis relationship means both substituents are on the same face (both up or both down); a trans relationship means opposite faces. Combine that with the axial/equatorial assignments and the table follows without memorisation.

4.16 Locked chairs and the tert-butyl group

A tert-butyl group is so bulky that it is essentially always equatorial; the axial conformer is about 20 kJ mol^{-1} higher and is not populated. This **locks** the chair: the ring can no longer flip, because the flip would put the tert-butyl group axial. Consequences: cis-4-tert-butyl-1-methylcyclohexane has methyl axial (it cannot be equatorial in the same chair), and the molecule is effectively frozen in one conformation.

KEY RESULT
A tert-butyl group is a conformational lock. If a question asks for the most stable conformation of a molecule containing one, put it equatorial and do not flip.

4.17 Stability and heat of combustion

The more stable an isomer, the lower its heat of combustion (less energy released on burning to CO_2 and H_2O). Heats of combustion therefore give an experimental ranking of isomer stability: trans-1,2-dimethyl-cyclohexane (e,e) < cis (a,e) < trans (a,a) in energy, which is the same order as their heats of combustion. For cis/trans alkenes the same logic applies, but the differences are smaller.

TRAP
The table above assumes identical substituents. With different substituents, "cis" and "trans" do not by themselves decide stability — you must place the larger group equatorial. A cis isomer with the large group equatorial can beat a trans isomer that forces it axial.

NOTE
A-values (the energy cost of an axial substituent) are tabulated in sheet X-04: CH_3 ≈ 7.3, Et ≈ 7.5, i-Pr ≈ 9.0, t-Bu ≈ 20 kJ mol^{-1}. They are the quantitative version of the argument on this page.

Every substitution changes the configuration at the reacting carbon in a way that is fixed by the mechanism. Learn the four entries in this sheet and you can answer most reaction-stereochemistry questions without memorising a list of reactions.

5.1 SN2: backside attack, clean inversion

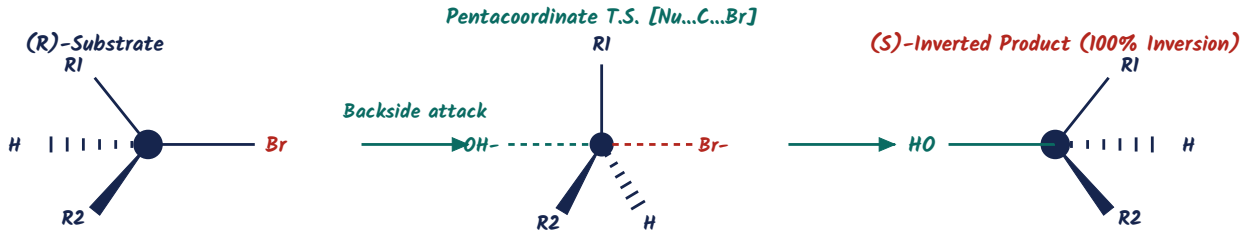


Fig. 5.1 — Walden inversion in S_N2: backside nucleophilic attack via a pentacoordinate transition state.

The nucleophile attacks the carbon from the side opposite the leaving group, through a trigonal-bipyramidal transition state, and the leaving group departs from the other side. The result is **inversion of configuration** — the Walden inversion. If the starting material is (R), the product is (S) (unless the priority order changes, which you must check).

$$\text{rate} = k[\text{substrate}][\text{nucleophile}]$$

stereochemistry: inversion (100 %)

- Favoured by methyl and primary halides, strong nucleophiles, polar aprotic solvents (DMSO, DMF, acetone).
- Retardation by β-branching; neopentyl halides react very slowly.
- A chiral secondary halide gives a chiral product with clean inversion; a racemic starting material gives a racemic product.

5.2 SN1: planar carbocation, racemisation

The leaving group departs first, giving a planar carbocation; the nucleophile can then attack either face. If the two faces are equally accessible, the product is racemic; in practice there is often slight inversion bias because the leaving group blocks one face for a moment. The result is **racemisation with partial inversion**, never clean retention.

$$\text{rate} = k[\text{substrate}]$$

stereochemistry: racemisation (with inversion bias)

- Favoured by tertiary halides, weak nucleophiles (water, alcohols), polar protic solvents (water, ethanol).
- Allylic and benzylic halides are intermediate; the carbocation is stabilised by resonance.
- A carbocation adjacent to a stereocentre can also racemise that centre by hydride or alkyl shifts.

KEY RESULT

S_N2 inverts, S_N1 racemises. The carbon skeleton does not rearrange in S_N2; it can in S_N1. If a product has a rearranged skeleton, the mechanism was S_N1 (or E_1).

5.3 Neighbouring-group participation and retention

If a nucleophile is already present in the molecule next to the leaving group (a carboxylate, a sulfide, an amide nitrogen), it can attack intramolecularly, displacing the leaving group with **inversion** at that step. The intermediate is then attacked by the external nucleophile at the same carbon, again with inversion. Two inversions give **net retention** of configuration.

retention = two inversions (double displacement)

5.4 Elimination

E_2 is a concerted anti-periplanar elimination: the base removes a β-H while the leaving group departs from the opposite side. The requirement is that H and L anti to each other, which fixes the stereochemistry of the alkene: the two groups that were anti in the conformer end up on opposite sides (trans) of the new double bond.

E_1 goes through the same carbocation as S_N1, so it loses stereochemical information at the reacting carbon and gives the more substituted (Zaitsev) alkene. In a rigid ring, E_2 needs a trans-diaxial arrangement of H and L; if the only β-H is not anti-periplanar, elimination is slow and substitution may dominate.

Mechanism	Stereochemical result	Stereospecific?	Example
S_N2	inversion	yes	primary halide + CN^-
S_N1	racemisation	no	tertiary halide + H_2O
E_2	anti elimination; alkene geometry set by the conformer	yes	2-bromobutane + EtO^-
E_1	Zaitsev alkene, lost stereochemistry at C-X	no	tertiary halide + heat
neighbouring group	retention (two inversions)	yes	mustard gas hydrolysis

Table 5.1 — the reaction table JEE asks for.

5.5 The substitution table JEE asks for

Substrate	Strong Nu^-, aprotic	Weak Nu, protic	Bulky base, heat
CH_3X	S_N2	S_N2 (slow)	—
primary RCH_2X	S_N2	S_N2 (slow)	E_2
secondary R_2CHX	$S_N2 + E_2$	$S_N1 + E_1$	E_2 (Zaitsev)
tertiary R_3CX	E_2	$S_N1 + E_1$	E_2 (Zaitsev)
allylic / benzylic	S_N2	S_N1	E_2

Table 5.2 — substrate plus conditions decides the mechanism.

TRAP
"Inversion" is about the position of the nucleophile relative to the leaving group, not about the R/S label. If the incoming group changes the CIP priority order, an inversion can leave the R/S descriptor unchanged. Work the priorities again after the reaction.

5.5b Three worked mechanism decisions

(a) 2-bromobutane + NaCN in DMSO. Secondary substrate, strong nucleophile, polar aprotic solvent → S_N2. The product is 2-methylbutanenitrile with **inversion** at C_2. If the starting material is (R)-2-bromobutane, the product is the (S) nitrile — but check the priority order, because CN now replaces Br and the ranking at C_2 changes.

(b) 2-bromo-2-methylbutane + water. Tertiary substrate, weak nucleophile, protic solvent → S_N1, and E_1 competes. The substitution product is racemic 2-methylbutan-2-ol; the elimination product is 2-methylbut-2-ene (Zaitsev). Heating favours the alkene; a stronger, bulkier base such as tert-butoxide gives almost entirely E_2.

(c) trans-1-bromo-4-tert-butylcyclohexane + ethoxide. E_2 requires a trans-diaxial H-C-C-Br arrangement. In this chair the Br is equatorial and the only β-H that is anti-periplanar is on C_2/C_6, which are equivalent; elimination gives 4-tert-butylcyclohexene. The tert-butyl group locks the chair, so there is no conformational escape: if the required anti-periplanar H were absent, elimination would be very slow.

Substrate + conditions	Mechanism	Stereochemical outcome
secondary + CN^-/DMSO	S_N2	inversion
tertiary + H_2O, warm	S_N1 / E_1	racemic substitution + Zaitsev alkene
secondary + EtO^-/EtOH, heat	E_2	anti elimination; alkene geometry from conformer
primary + $t\text{-}BuO^-$, heat	E_2	anti elimination, Hofmann product

Table 5.3 — three decisions, one method: substrate, nucleophile/base strength, solvent.

5.6 Syn and anti addition

Addition to a C=C can deliver the two new groups to the same face (**syn**) or opposite faces (**anti**). The stereochemistry of the product is fixed by the mechanism, not by the reagent's name:

- **Syn**: H_2/Pd (both H from the same face), hydroboration-oxidation (B and H syn), dihydroxylation with OsO_4 (both OH syn), hydrogenation of an alkene.
- **Anti**: Br_2 (bromonium ion opened from the back), HBr (via the bromonium/carbocation), epoxidation followed by anti opening in base.

Alkene	syn addition gives	anti addition gives
cis-2-butene	meso-2,3-butanediol / meso dibromide	racemic pair
trans-2-butene	racemic pair	meso compound
cyclohexene	cis-1,2 product	trans-1,2 product
1-methylcyclohexene	cis (both on one face)	trans

Table 5.4 — the same reagent can give meso or racemic depending on the alkene geometry. This is the most examined table in the chapter.

KEY RESULT
cis alkene + syn addition → meso. cis alkene + anti addition → racemic. trans alkene + syn addition → racemic. trans alkene + anti addition → meso. Reverse the alkene or the addition and you reverse the outcome.

5.7 Hydrogenation of an optically active alkene

If the alkene already contains a stereocentre, hydrogenation creates a **new** stereocentre and can give two diastereomers, in unequal amounts (diastereoselective addition). The existing stereocentre biases which face of the alkene is hydrogenated, so the product is not a racemic mixture: it is a mixture of diastereomers, and the optical activity is retained because the original centre is untouched.

If the alkene is the only source of chirality (for example an alkene that is chiral only because of the arrangement of its substituents), hydrogenation destroys the chirality and the product is optically inactive. This is the sense in which "hydrogenation of an optically active alkene can destroy the activity".

5.8 Addition to a C=C that already has stereocentres

When a new centre is created next to an existing one, the two possible products are diastereomers. The facial selectivity is decided by steric hindrance from the existing substituents (the "Cram" argument, sheet X-06): the reagent approaches the face that is less crowded. In a ring, the approach is usually from the less hindered face, which is why substituents direct hydrogenation cis or trans.

5.9 Addition to a carbonyl

Nucleophilic addition to a carbonyl creates a stereocentre when the two groups on the carbonyl carbon are different. With an achiral carbonyl and an achiral nucleophile, the two faces are equivalent, so a racemic mixture results. With a chiral carbonyl (one that already contains a stereocentre) the faces are diastereotopic, giving diastereomers — the basis of the Cram and Felkin-Anh rules.

5.10 A worked addition problem

cis-2-Butene + Br_2. Anti addition of Br_2 gives the (2R,3R)/(2S,3S) pair — a racemic mixture.
trans-2-Butene + Br_2. The same anti addition now gives the (2R,3S) meso compound. Check by drawing the bromonium ion on one face and opening it from the other: the relative positions of the two Br atoms invert with the alkene geometry.

TRAP
Do not write "addition of Br_2 gives the trans product". It gives the anti addition; whether the final molecule is cis or trans, meso or racemic, depends on the alkene. Draw the bromonium ion every time — it is faster than remembering four cases.

5.10b Counting the products of an addition

Addition questions usually ask for the number of products and whether they are optically active. Work in three steps: decide syn or anti from the reagent, draw the product for one face, then check the product for symmetry.

1. **cis-2-Butene + H_2/Pd (syn)**: both H on one face → the two CH_3 groups end up on the same side of the new C-C bond → the meso (2R,3S) pair collapses to one achiral compound: 1 product.
2. **trans-2-Butene + H_2/Pd (syn)**: the CH_3 groups end up on opposite sides → (2R,3R) and (2S,3S) → a racemic pair: 2 products, optically inactive as a mixture.
3. **cis-2-Butene + Br_2 (anti)**: the two Br add from opposite faces → (2R,3R)/(2S,3S) → racemic pair: 2 products.
4. **trans-2-Butene + Br_2 (anti)**: opposite faces twice → the meso (2R,3S) compound: 1 product.

Alkene	Reagent (mode)	Product	Active?
cis-2-butene	H_2/Pd (syn)	meso-2,3-dimethylbutane-like — one compound	no
trans-2-butene	H_2/Pd (syn)	racemic pair	mixture inactive
cis-2-butene	Br_2 (anti)	racemic pair	mixture inactive
trans-2-butene	Br_2 (anti)	meso dibromide	no
cyclohexene	Br_2 (anti)	trans-1,2-dibromocyclohexane (racemic)	mixture inactive
cyclohexene	OsO_4/H_2O_2 (syn)	cis-1,2-cyclohexanediol (meso)	no

Table 5.5 — the same two reagents, four different outcomes, decided by alkene geometry and addition mode.

KEY RESULT
The number of products is decided by symmetry, not by the reagent. Ask: after the addition, is there a plane? If yes, one compound; if no, a racemic pair.

TRAP
A racemic pair is "optically inactive" as a mixture but each molecule is chiral. An answer that says "the product is achiral because it is racemic" is wrong; say "a racemic mixture, therefore optically inactive".

5.11 The checklist

1. Write the molecular formula and compute the degree of unsaturation.
2. Enumerate the **structural** skeletons first (chains, then rings, then multiple bonds).
3. For each skeleton, mark every geometrical centre and every stereocentre.
4. Count the stereoisomers skeleton by skeleton; never add n's across skeletons.
5. Report structural, geometrical and optical counts separately, then the total.

5.12 Worked: $C_5H_{11}Cl$

Degree of unsaturation = 0, so the compound is acyclic and saturated (or a ring). Eight structural isomers exist; the stereochemistry multiplies some of them.

Skeleton	Structural isomers	Stereoisomers	Optically active?
n-pentane chain	3 (1-, 2-, 3-chloro)	1 + 2 + 2 = 5	2-chloro and 3-chloro (each a pair)
2-methylbutane	4 (1-, 2-, 3-, 4-chloro)	1 + 2 + 1 + 1 = 5	2-chloro only
2,2-dimethylpropane	1 (1-chloro)	1	no
cyclopentane	1	1	no
methylcyclobutane	2	2 + 2	the 1-chloro-1-methyl isomer is achiral
dimethylcyclopropanes	4	several	check each for a plane

Table 5.6 — the enumeration skeleton by skeleton. Add the columns only after each row is justified.

The exam-safe answer is to draw each skeleton, mark the stereocentres, and state the count for each. A single total with no working is the commonest way to lose marks on this question, even when the number is right.

5.13 Worked: dichloro derivatives of butane

$C_4H_8Cl_2$ has two structural families: the butane chain and the 2-methylpropane skeleton. On the butane chain the chlorines can be 1,1; 1,2; 1,3; 1,4; 2,2; 2,3. Of these, 1,2 and 2,3 create stereocentres, and 2,3 also gives a meso form; 1,3 has no stereocentre but can be chiral if the two ends differ (they do not here). On the branched skeleton, 1,1; 1,2; 1,3 and 2,2 are possible, with 1,2 giving a stereocentre.

KEY RESULT
Enumerate skeletons → place substituents → mark centres → test symmetry. Four steps, in that order, for every counting question. The formula sheet (J-29) is only a check on the result.

5.14 Separable fractions

A reaction that creates a stereocentre in an achiral environment gives a **racemic mixture**, which is optically inactive and separable only by resolution. A reaction that creates a second stereocentre in a molecule that already has one gives **diastereomers**, which are separable by ordinary methods. This single distinction decides whether the phrase "the product is optically inactive" is a statement about the mechanism (racemic) or about the substrate (meso).

TRAP
A meso product and a racemic product both rotate zero. The question usually tells you which one is being asked about by giving the alkene geometry or the substrate's chirality — read that first.

These are the twenty statements that turn a correct idea into a wrong answer. Each is written as the trap, followed by the correction.

1. "Geometrical isomerism needs a C=C." **No** — a ring, a C=N or an N=N centre also works.
2. "A molecule with two stereocentres is always chiral." **No** — it may be meso.
3. "A molecule with no stereocentre is achiral." **No** — allenes, biphenyls, helicenes.
4. "Enantiomers have different boiling points." **No** — they differ only in the sign of rotation.
5. "Diastereomers are mirror images." **No** — that is the definition of enantiomers.
6. "A racemic mixture is achiral." **No** — it contains chiral molecules and is optically inactive by external compensation.
7. "Meso compounds rotate light." **No** — internal compensation, $\alpha = 0$.
8. "cis is always more polar." Usually, but check the vectors; a cis isomer with cancelling moments can have $\mu = 0$.
9. "trans always has the higher boiling point." **Usually not** — the cis isomer is more polar. Maleic/fumaric is the exception for melting point, not a rule for boiling point.
10. "A 90° rotation of a Fischer projection is allowed." **No** — it inverts every centre.
11. "Swapping two groups twice changes the configuration." **No** — two swaps restore it.
12. "S_N1 always gives exactly 50:50." **No** — racemisation with inversion bias.
13. "S_N2 changes the R/S label." Not necessarily — the priority order can change.
14. "E_2 gives the trans alkene." It gives the anti-periplanar product; the geometry follows from the conformer.
15. "Br_2 addition gives the trans product." It gives anti addition; meso or racemic depends on the alkene.
16. "Hydrogenation always destroys optical activity." Only if the chirality depended on the alkene.
17. "D and L mean (+) and (−)." **No** — independent labels.
18. "More stereocentres means more isomers." Only up to symmetry; meso forms reduce the count.
19. "A stereocentre must be carbon." **No** — N, P, S and metal centres can be stereogenic.
20. "If the mirror image looks the same, the compound is achiral." Only if it is superimposable — "looks the same" after a rotation is exactly the test, but a reflection is not a rotation.

KEY RESULT

Most of these traps are the same mistake in different clothes: a rule about one example is being applied to every example. When a question looks familiar, check that the symmetry has not changed.

Definitions in one line each

- **Stereoisomers**: same connectivity, different arrangement in space.
- **Enantiomers**: non-superimposable mirror images; identical physical properties, opposite rotation.
- **Diastereomers**: stereoisomers that are not mirror images; different physical properties.
- **Meso**: achiral despite stereocentres, because of an internal plane or centre.
- **Racemic**: 1:1 mixture of enantiomers; inactive by external compensation.
- **Chirality**: the molecule is not superimposable on its mirror image.
- **Stereocentre**: an atom whose substituents can be interchanged to give a stereoisomer.
- **Optical purity / ee**: the fraction of the mixture that is one enantiomer, expressed as a percentage.

The formulas

$$[\alpha] = \alpha / (l \times c) \quad l \text{ in dm, } c \text{ in g mL}^{-1}$$

$$\text{ee} = (\text{observed } [\alpha] / [\alpha] \text{ of pure enantiomer}) \times 100\ \%$$

$$\text{geometrical: } 2^n \text{ (dissimilar ends); } (2^n + 2^{\text{ceil}(n/2)})/2 \text{ (similar ends)}$$

optical: see Table 3.12 (J-19)

The symmetry test

- σ present → achiral. i present → achiral. S_n (n even) → achiral.
- One stereocentre → always chiral.
- Two or more → test; symmetric chains may be meso.
- No stereocentre → still test for axial/planar chirality.

The projection conventions

- Fischer: vertical back, horizontal forward; 180° rotation allowed, 90° forbidden.
- Newman: front carbon = point, back = circle; θ = 0° eclipsed, 60° staggered.
- Sawhorse: side view; keep the viewing direction when converting.
- Wedge-dash: wedge = towards you, hash = away.

The reaction rules

- S_N2: inversion, second-order, aprotic, primary.
- S_N1: racemisation (with bias), first-order, protic, tertiary.
- E_2: anti-periplanar, stereospecific, Zaitsev with small bases.
- cis alkene + syn addition → meso; + anti → racemic.
- trans alkene + syn → racemic; + anti → meso.

The conformational ladder

- anti > gauche > eclipsed (butane); chair > twist-boat > boat (cyclohexane).
- Flip: a ⇄ e, up stays up.
- 1,2-cis = a,e; 1,3-cis = e,e; 1,4-cis = a,e (identical substituents).
- tert-Butyl locks the chair equatorial.

KEY RESULT
Draw the molecule, find the symmetry, then name it. Every question in this chapter is a version of that sequence.

Numbers worth carrying into the exam

Quantity	Value	Where it is used
C=C rotation barrier	250–270 kJ mol^{-1}	why geometrical isomers exist
C–C rotation barrier (ethane)	≈ 12.5 kJ mol^{-1}	conformers interconvert
gauche CH_3/CH_3 interaction	≈ 3.8 kJ mol^{-1}	butane populations
chair → twist-boat	≈ 23 kJ mol^{-1}	chair is the minimum
cyclohexane ring flip	≈ 45 kJ mol^{-1}	fast at 25 °C
A-value of CH_3	≈ 7.3 kJ mol^{-1}	≈ 95 % equatorial
A-value of t-Bu	≈ 20 kJ mol^{-1}	conformational lock
2,2′-disubstituted biphenyl barrier	80–160 kJ mol^{-1}	atropisomers isolable

Table R.1 — orders of magnitude, not measurements.

The five-second checks

1. Does the molecule have a plane or a centre? If yes → achiral, stop.
2. Are the two ends of the chain identical? If yes → use the symmetric counting row, not 2^n.
3. Is the lowest-priority group pointing towards you? If yes → reverse the R/S answer.
4. Is the nucleophile strong and the solvent aprotic? If yes → S_N2 (inversion).
5. Did the alkene geometry change the meso/racemic outcome? Draw the bromonium ion, do not recall a rule.

KEY RESULT
Six ideas carry the whole chapter: rotation can be blocked; two different groups at each end; symmetry decides activity; a plane or centre decides achirality; inversion for S_N2 and racemisation for S_N1; equatorial beats axial.